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K-COHOMOLOGY OF SEVERI-BRAUER VARIETIES AND THE NORM RESIDUE HOMOMORPHISM

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ABSTRACT. The basic purpose of this paper is to prove bijectivity of the norm residue homomorphism $R_{F,n}$: $K_2(F)/nK_2(F) \to H^2(F, \mu_n^{\otimes 2})$ for any field F of characteristic prime to n. In particular, if $\mu_n \subset F$, then any central simple algebra of exponent n is similar to a tensor product of cyclic algebras. In the course of the proof we obtain partial degeneracy of the Gersten spectral sequence, and we compute some K-cohomology groups of Severi-Brauer groups corresponding to cyclic algebras of prime degree. The fundamental theorem also gives us several corollaries.

Bibliography: 27 titles.

Introduction

This paper is devoted to the study of the norm residue homomorphism [26]

$$R_{F,n}\colon K_n(F)/nK_2(F) \to H^2\big(F, \mu_n^{\otimes 2}\big).$$

The fundamental theorem, which will be proved in §§9–11, states that $R_{F,n}$ is always an isomorphism. In the case $n = 2^r$ this theorem was proved by Merkur'ev in [1]. That proof used computations of the groups $\ker R_{F,n}$ and $\operatorname{coker} R_{F,n}$ for F a field of rational functions on a conic, which were obtained by Suslin in [6]. To a large extent the present paper was inspired by Merkur'ev's work in [1], which showed that the study of Severi-Brauer varieties must play a central role in the investigation of the norm residue homomorphism. It was this work which was the point of departure for our subsequent investigations. However, Merkur'ev's method did not work in the case of a prime $n = p \neq 2$: in this situation it was necessary to prove that, if $D = A_\xi(a, b)$ is a cyclic algebra of degree p over the field F (see [3]), and if $c \in F^*$ is an element such that $\{a, b, c\} \in p \cdot K_3^M(F)$, then c is a reduced norm for the algebra D (in the case $p = 2$ this assertion is a consequence of the Arason-Pfister theorem about quadratic forms). Here we shall use another method, based on the computation of certain K-cohomology groups of Severi-Brauer varieties, which will be given in §8. This computation makes use of computations of Quillen in the K-theory of Severi-Brauer varieties, the Gersten spectral sequence, and the Riemann-Roch formula for higher K-groups. In order to prove that $K_2(F) \cong H^0(X, \mathcal{K}_2)$, we use some results on K_2 of division algebras of squarefree degree (i.e., whose degree is not divisible by the square

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of a prime), which will be obtained in §§4–7. In §12 we derive the result mentioned above concerning the reduced norms in cyclic algebras as a consequence of the fundamental theorem. One of the most interesting applications of our methods is a Hilbert Theorem 90 for K_2 which describes the kernel of the transfer map $K_2(E) \overset{N_{E/F}}{\to} K_2(F)$ for a cyclic extension E/F; it will be proved in §§13 and 14. In §17 we use this Hilbert Theorem 90 and the existence of the reduced norm for K_2 in computing K_2 of division algebras over local and global fields. Other applications of the fundamental theorem are contained in §§15, 16 and 18.

The material in §§4–8, 12, 15–18, and also the method used below to prove the fundamental theorem, belong to A. A. Suslin. The Hilbert Theorem 90 for quadratic extensions was proved by A. S. Merkur'ev, and it is his method that is used below to prove the theorem in the case of cyclic extensions of prime degree (§13). The general case of this theorem was proved by Suslin (§14), and it was his observation that the equality ${}_nK_2(F) = \{\mu_n, F^*\}$ follows from the Hilbert Theorem 90.

The basic results of this paper were announced in [2].

§1. The homomorphism θ_L

(1.1) Let D be a central simple algebra over a field F, and let L/F be a splitting field for D of finite degree. The canonical homomorphisms $\theta_L\colon K_i(L) \to K_i(D)$ can be defined in any of the following equivalent ways:

a) θ_L coincides with the composition

$$K_i(L) = K_i(M_n(L)) = K_i(D_L) \overset{N_{D_L/D}}{\to} K_i(D).$$

b) L can be imbedded as a maximal subfield in an algebra D' similar to D: $L \overset{\varphi}{\hookrightarrow} D' \sim D$, and θ_L coincides with the composition

$$K_i(L) \to K_i(D') = K_i(D).$$

c) If V denotes a simple D_L-module, then the homomorphism θ_L is induced by the exact functor

$$L\text{-mod} \Rightarrow D\text{-mod}\colon E \Rightarrow V \otimes_L E.$$

(1.2) LEMMA. *The homomorphisms θ_L have the following properties*:

(1.2.1) *If $E \supset L$ are splitting fields for the algebra D, then the diagram*

$$\begin{array}{ccc} K_i(E) & \overset{N_{E/L}}{\to} & K_i(L) \\ {\scriptstyle \theta_E}\searrow & & \swarrow{\scriptstyle \theta_L} \\ & K_i(D) & \end{array}$$

is commutative.

(1.2.2) *If $\sigma \in \mathrm{Aut}(L/F)$, then the diagram*

$$\begin{array}{ccc} K_i(L) & \overset{\sigma_*}{\to} & K_i(L) \\ {\scriptstyle \theta_L}\searrow & & \swarrow{\scriptstyle \theta_L} \\ & K_i(D) & \end{array}$$

is commutative.

(1.2.3) *The following diagrams are commutative*:

$$\begin{array}{ccc} & K_i(L) & \\ {\scriptstyle\theta_L}\swarrow & & \searrow{\scriptstyle\theta_L} \\ K_i(D) & \cong & K_i(M_n(D)) \end{array}$$

(1.2.4) *If* $L \supset E \supset F$, *and* L *splits* D, *and hence* D_E, *then the diagram*

$$\begin{array}{ccc} & K_i(L) & \\ {\scriptstyle\theta_L}\swarrow & & \searrow{\scriptstyle\theta_L} \\ K_i(D_E) & \overset{N_{D_E/D}}{\to} & K_i(D) \end{array}$$

is commutative.

(1.2.5) *Let* F'/F *be an arbitrary extension,*

$$(L \otimes_F F')/\mathrm{Rad}(L \otimes_F F') = \prod_{j=1}^{m} L_j$$

and s_j *the length of the local factor in* $L \otimes_F F'$ *corresponding to* L_j. *Then the following diagram is commutative*:

$$\begin{array}{ccc} K_i(L) & \overset{\theta_L}{\to} & K_i(D) \\ \downarrow & & \downarrow \\ \prod_{j=1}^{m} K_i(L_j) & \overset{\Sigma s_j\theta_{L_j}}{\to} & K_i(D_{F'}). \end{array}$$

The proof of the lemma is trivial, and we shall omit it.

For an arbitrary central simple algebra D/F we set

$$\overline{K}_i(D) = \mathrm{coker}\left(\prod_{\substack{L \text{ a splitting} \\ \text{field for } D}} K_i(L) \to K_i(D) \right).$$

The next fact is a consequence of the above properties of the homomorphisms θ_L.

(1.3) LEMMA. a) *If* F'/F *is an arbitrary extension, then the natural map* $K_2(D) \to K_2(D_{F'})$ *induces a homomorphism* $\overline{K}_2(D) \to \overline{K}_2(D_{F'})$.

b) *If the extension* F'/F *is finite, then* $N_{D_{F'}/D}$ *induces a map* $\overline{K}_2(D_{F'}) \to \overline{K}_2(D)$, *which we shall also denote by* $N_{D_{F'}/D}$.

(1.4) COROLLARY. *Let* n *denote the degree of the algebra* D/F (*i.e., the square root of its dimension*). *Then*:

a) $n\overline{K}_2(D) = 0$.

b) *If* $([F': F], n) = 1$, *then* $\overline{K}_2(D) \hookrightarrow \overline{K}_2(D_{F'})$.

§2. Specialization

If O is a discrete valuation ring with field of fractions F, residue field k, and local parameter π, then there exists a canonical homomorphism s_π: $K_2(F) \to K_2(k)$ defined by

$$s_\pi(\{x, y\}) = \left\{ \left(\frac{x}{\pi^{\mathrm{ord}\, x}} \right), \left(\frac{y}{\pi^{\mathrm{ord}\, y}} \right) \right\}$$

(see [11]). Now let O be a regular local ring, let F be the field of fractions and k the residue field of O, and let $\pi = (\pi_1, \ldots, \pi_r)$ be a regular system of parameters. If we let F_i denote the field of fractions of the ring $O/(\pi_1, \ldots, \pi_i)$, then, by iterating the maps constructed above, we obtain the homomorphism

$$s_\pi: K_2(F) \xrightarrow{s_{\pi_1}} K_2(F_1) \xrightarrow{s_{\pi_2}} K_2(F_2) \to \cdots \xrightarrow{s_{\pi_r}} K_2(F_r) = K_2(k).$$

The map s_π is called the specialization homomorphism relative to the system π. It clearly depends on the choice of π and the ordering of the parameters π_i. Nevertheless, its restriction to $K_2(O)$ coincides with the natural homomorphism $K_2(O) \to K_2(k)$ induced by the map $O \to k$.

The homomorphism s_π has the following trivial properties (which are proved by reducing to the case of a discrete valuation ring and then making a direct computation).

(2.1.1) If O' is etale over O, then the diagram

$$\begin{array}{ccc} K_2(F) & \to & K_2(F') \\ \downarrow s_\pi & & \downarrow s_\pi \\ K_2(k) & \to & K_2(k') \end{array}$$

is commutative.

(2.1.2) Suppose that A/O is finite etale, and A is an integral domain. Let $\mathfrak{M}_1, \ldots, \mathfrak{M}_r$ be maximal ideals of A, and let $k_1, \ldots, k_r$ be the corresponding residue fields. Let L denote the field of fractions of A; then the diagram

$$\begin{array}{ccc} K_2(L) & \to & K_2(F) \\ \downarrow (s_{A_{\mathfrak{M}_i}}, \pi) & & \downarrow s_\pi \\ \prod K_2(k_i) & \to & K_2(k) \end{array}$$

is commutative.

(2.1.3) Under the conditions of (2.1.2), suppose further that $\operatorname{Spec} A \to \operatorname{Spec} O$ is a Galois covering with group G. Then the map s_π: $K_2(L) \to \prod K_2(k_i) = K_2(A/\pi A)$ is G-equivariant.

(2.2) Later we shall also use specialization homomorphisms in the noncommutative case.

(2.2.1) Suppose that O is a regular local ring in an algebraic domain, and Λ is an Azumaya algebra over O. By means of a minor modification of Quillen's arguments [20], we see that the following sequence is exact:

$$0 \to K_i(\Lambda) \to K_i(\Lambda_F) \xrightarrow{\partial} \prod_{\operatorname{ht}\mu=1} K_{i-1}(\Lambda(\mu)) \xrightarrow{\partial} \prod_{\operatorname{ht}\mu=2} K_{i-2}(\Lambda(\mu)) \xrightarrow{\partial} \cdots.$$

In particular, if O is a discrete valuation ring, then we have the exact sequence

$$0 \to K_i(\Lambda) \to K_i(\Lambda_F) \to K_{i-1}(\overline{\Lambda}) \to 0.$$

(2.2.2) LEMMA. *There exists a unique homomorphism* s_π: $K_2(\Lambda_F) \to K_2(\overline{\Lambda})$ *such that*
a) *the restriction* $s_\pi|_{K_2(\Lambda)}$ *is induced by the ring homomorphism* $\Lambda \to \overline{\Lambda}$, *and*
b) $s_\pi(\{\pi, \lambda\}) = 0$ *for any* $\lambda \in \Lambda^*$.

(2.2.3) Suppose that $\overline{\Lambda}$ is a division algebra. Then every nonzero element in the algebra Λ_F can be written in the form $x = \pi^{\operatorname{ord} x} y$, where $y \in \Lambda^*$; in particular, Λ_F is also a

division algebra. Hence, we can make use of Rehmann's representation of the group $K_2(\Lambda_F)$ (see [21]). It is easy to verify that, if $x_i, y_i \in \Lambda_F^*$, $\prod[x_i \text{ and } y_i] = 1$, then

$$s_\pi(c(x_1, y_1)\cdot\cdots\cdot c(x_n, y_n)) = c\left(\overline{\frac{x_1}{\pi^{\operatorname{ord} x_1}}}, \overline{\frac{y_1}{\pi^{\operatorname{ord} y_1}}}\right)\cdot\cdots\cdot c\left(\overline{\frac{x_n}{\pi^{\operatorname{ord} x_n}}}, \overline{\frac{y_n}{\pi^{\operatorname{ord} y_n}}}\right).$$

(2.2.4) Properties (2.1.1) and (2.1.2) are preserved in the present situation.

(2.2.5) Suppose that O' is a finite etale O-algebra which splits the algebra Λ. Then the diagram

$$\begin{array}{ccc} K_2(F') & \xrightarrow{\theta_{F'}} & K_2(\Lambda_F) \\ \downarrow s_\pi & & \downarrow s_\pi \\ K_2(\overline{O}') & \xrightarrow{\theta_{\overline{O}'}} & K_2(\overline{\Lambda}) \end{array}$$

is commutative.

§3. The regular representation

Let F be an arbitrary field, and let D/F be a central simple algebra of squarefree degree.

(3.1) PROPOSITION. *There exist polynomial mappings (with respect to some basis of D/F) $\varphi_i, \psi_i: D \to D$ with the following properties*:

a) $\varphi_i(1), \psi_i(1) \in D^*$.

b) *If $x \in \mathrm{SL}_1(D)$ and $\varphi_i(x), \psi_i(x) \in D^*$, then*

$$x = [\varphi_1(x), \psi_1(x)]\cdot\cdots\cdot[\varphi_r(x), \psi_r(x)].$$

PROOF. Since the group $\mathrm{SK}_1(D_E)$ is trivial for any extension E/F (see [4] and [27]), it follows that we can find polynomial mappings φ_i' and ψ_i' such that at the generic point $x \in \mathrm{SL}_1(D)$ the elements $\varphi_i'(x)$ and $\psi_i'(x)$ are invertible and

$$x = [\varphi_1'(x), \psi_1'(x)]\cdot\cdots\cdot[\varphi_k'(x), \psi_k'(x)].$$

Since the rational points are dense in $\mathrm{SL}_1(D)$, we can find a rational point $x_0 \in \mathrm{SL}_1(D)$ such that $\varphi_i'(x_0), \psi_i'(x_0) \in D^*$. We choose an equality $x_0^{-1} = [\alpha_1, \beta_1]\cdot\ldots\cdot[\alpha_l, \beta_l]$; then we have a new representation for x which satisfies both of our conditions:

$$x = xx_0x_0^{-1} = [\varphi_1'(xx_0), \psi_1'(xx_0)]\cdot\cdots\cdot[\varphi_k'(xx_0), \psi_k'(xx_0)]\cdot[\alpha_1, \beta_1]\cdot\cdots\cdot[\alpha_l, \beta_l],$$

i.e., we can set $\varphi_1(x) = \varphi_1'(xx_0), \ldots, \varphi_k(x) = \varphi_k'(xx_0)$, $\varphi_{k+1}(x) = \alpha_1, \ldots, \varphi_{k+l}(x) = \alpha_l$, $\psi_1(x) = \psi_1'(xx_0), \ldots$.

(3.2) Under the conditions of (3.1), suppose that $\varphi_i(x), \psi_i(x) \in D^*$. Then we can set

$$c(x) = c(\varphi_1(x), \psi_1(x))\cdot\cdots\cdot c(\varphi_r(x), \psi_r(x))\cdot c(\psi_r(1), \varphi_r(1))$$
$$\cdot\cdots\cdot c(\psi_1(1), \varphi_1(1)) \in \mathrm{St}_1(D),$$

i.e., we have a "rational map" from $\mathrm{SL}_1(D)$ to $\mathrm{St}_1(D)$, which is defined at the point 1 and takes it to 1. It is also clear that c is a section of the natural projection $\mathrm{St}_1(D) \to \mathrm{SL}_1(D)$.

(3.3) PROPOSITION. *Suppose that O is a discrete valuation ring, F is its field of fractions, k is its residue field, π is a local parameter of O, Λ is an Azumaya algebra over O, and $D = \Lambda_F$. Further suppose that $\overline{\Lambda}$, and hence also D, are division algebras. Then there exist polynomial mappings $\varphi_i, \psi_i: D \to D$ that satisfy the following conditions*:

a) *Their coefficients (with respect to some basis of Λ/O) lie in O.*

b) *If* $x \in \mathrm{SL}_1(D)$ *and* $\varphi_i(x), \psi_i(x) \in D^*$, *then*

$$x = [\varphi_1(x), \psi_1(x)] \cdot \cdots \cdot [\varphi_r(x), \psi_r(x)].$$

c) $\varphi_i(1), \psi_i \in \Lambda^*$.

PROOF. It is easy to construct maps φ_i' and ψ_i' satisfying a) and b). We can further assume that $\overline{\varphi_i'}|_{\mathrm{SL}_1(\bar{\Lambda})} \not\equiv 0$ and $\overline{\psi_i'}|_{\mathrm{SL}_1(\bar{\Lambda})} \not\equiv 0$. In fact, suppose that $\overline{\varphi_i'}|_{\mathrm{SL}_1(\bar{\Lambda})} \equiv 0$. Choose a basis of Λ/O; then φ_i' will be given by a set of polynomials $\varphi_i'^{kl}$ with coefficients in O. Our supposition is equivalent to the inclusion $\varphi_i'^{kl} \in (\pi, \mathrm{Nrd} - 1)$, i.e. $\varphi_i' = \varphi_i'' \cdot \pi + (\mathrm{Nrd} - 1)\chi_i$, and we can replace φ_i' by φ_i''. But this process cannot be continued indefinitely. Thus, there exists an element $x_0 \in \mathrm{SL}_1(\Lambda)$ such that $\varphi_i'(x_0), \psi_i'(x_0) \in \Lambda^*$, and now the proof can be finished in the same way as before (note that $\mathrm{SL}_1(\Lambda) = [\Lambda^*, \Lambda^*]$).

(3.4) REMARK. If the maps φ_i and ψ_i satisfy the conditions of Proposition (3.3) and $x \in \mathrm{SL}_1(D) \cap (1 + \pi\Lambda)$, then $\varphi_i(x), \psi_i(x) \in \Lambda^*$, and, consequently, the element $c(x)$ is defined. Moreover, $c(x) \in \mathrm{St}_1(\Lambda)$ and $\overline{c(x)} = 1$.

§4. The group $K_2(D(T))$

If D is a central simple F-algebra, then there exists a split exact sequence

$$0 \to K_2(D) \to K_2(D(T)) \xrightarrow{\partial} \prod_{\mu \in \mathrm{Max}\, F[T]} K_1(D(\mu)) \to 0.$$

We shall later need some formulas for computing ∂.

(4.1) LEMMA. *Let the conditions of Proposition* (3.3) *be fulfilled.*

a) *If* $\lambda \in \Lambda^*$, *then* $\partial(\{\pi, \lambda\}) = \bar{\lambda} \pmod{[\bar{\Lambda}^*, \bar{\Lambda}^*]}$.

b) *Suppose that* $\bar{\Lambda} = M_n(k)$, *and let* $\lambda \in \Lambda$ *be a generator of a maximal right ideal* (*thus*, $\Lambda/\lambda\Lambda \cong k^n$). *If we identify* $K_1(\bar{\Lambda})$ *with* $K_1(k) = k^*$, *then* $\partial(\{\lambda, x\}) = \bar{x}$ *for any* $x \in O^*$.

(4.2) PROPOSITION. *Suppose that* D/F *is a central simple algebra of prime degree* p, *and the field* F *does not have extensions of degree prime to* p. *Then the group* $K_2(D(T))$ *is generated by the elements in* $K_2(D)$ *and the symbols* $\{\lambda\ x\}$, *where* $\lambda \in D[T]$ *and* $x \in F[T]$.

PROOF. Let L denote the subgroup of $K_2(D(T))$ generated by $K_2(D)$ and the elements $\{\lambda, x\}$ of the indicated form. It suffices to show that for any $\mu \in \mathrm{Max}(F[T])$ we have $\partial(L) \supset K_1(D(\mu))$. The proof proceeds by induction on the degree $[F(\mu): F] = p^k$.

a) The case $k = 0$ is trivial.

b) Suppose that we have proved our claim for all $k < k'$, and let $[F(\mu): F] = p^k$. First suppose that the field $F(\mu)$ splits the algebra D. We consider the natural projection $D[T] \to D(\mu) = M_p(F(\mu))$, and we choose a generator $\lambda \in D[T]$ of the preimage of the maximal ideal of the algebra $D(\mu)$. This can be done, since $D[T]$ is a principal ideal ring. It is easy to see that $\mu = (\mathrm{Nrd}\, \lambda)$, and hence λ does not have any zeros other than μ. If we choose a representative $x \in F[T]$ of degree less than p^k for any $\bar{x} \in F(\mu)^*$, then

1) $\partial_\mu(\{\lambda, x\}) = \bar{x}$, and

2) $\partial_\nu(\{\lambda, x\}) = 1$, if $[F(\nu): F] \geqslant p^k$ and $\nu \neq \mu$.

c) Finally, we suppose that $[F(\mu): F] = p^k$, and $F(\mu)$ is not a splitting field for the algebra D. Then $\mu = (f)$, and $\deg f = p^k$. If we choose a representative $\lambda \in D[T]$ of degree less than p^k for any element $\bar{\lambda} \in D(\mu)^*$, then

1) $\partial_\mu(\{f, \lambda\}) = \lambda$, and

2) $\deg(\mathrm{Nrd}\, \lambda) < p \cdot p^k = p^{k+1}$.

Hence λ has no zeros at places of degree greater than p^k; thus, $\partial_v(\{f, \lambda\}) = 1$, if $[F(v): F] > p^k$.

3) $\partial_v(\{f, \lambda\}) = 1$, if $[F(v): F] = p^k$, $F(v)$ is not a splitting field of D, and $v \neq \mu$.

In fact, $\lambda \pmod{v \cdot D[T]}$ is not a zero, since $\deg \lambda < p^k$; on the other hand, $D[T]/vD[T] = D(v)$ is a division algebra, and so the element $\lambda \pmod{vD[T]}$ is invertible.

(4.3) COROLLARY. *Under the conditions of* (4.2), *for any* $u \in K_2(D(T))$ *the element* $s_{T-1}(u) - s_T(u)$ *lies in the subgroup of* $K_2(D)$ *generated by the symbols* $\{\lambda, x\}$, $\lambda \in D^*$, $x \in F^*$.

§5. Triviality of $\bar{K}_2$ for algebras of squarefree degree

(5.1) PROPOSITION. *Suppose that* D/F *is a division algebra of prime degree* p, *and the field* F *does not have extensions of degree prime to* p. *Then the group* $K_2(D)$ *is generated by the symbols* $\{\lambda, x\}$, *where* $\lambda \in D^*$ *and* $x \in F^*$.

PROOF. We shall use Rehmann's description of $K_2(D)$ (see [21]). We choose polynomial mappings $\varphi_1, \psi_1, \ldots, \varphi_r, \psi_r$ which satisfy the conditions (3.1). Every element of $K_2(D)$ has the form $u = c(\sigma_1, \tau_1) \cdot \cdots \cdot c(\sigma_n, \tau_n)$, where $\sigma_i, \tau_i \in D^*$ and $[\sigma_1, \tau_1] \cdot \cdots \cdot [\sigma_n, \tau_n] = 1$. We set

$$\begin{aligned}\sigma &= \big[\sigma_1 + T(1-\sigma_1), \tau_1 + T(1-\tau_1)\big] \cdot \cdots \cdot \big[\sigma_n + T(1-\sigma_n), \tau_n + T(1-\tau_n)\big] \\ &\equiv \mathrm{SL}_1(D(T)),\end{aligned}$$

$$\begin{aligned}v = c(&\sigma_1 + T(1-\sigma_1), \tau_1 + T(1-\tau_1)) \\ &\cdot \cdots \cdot c(\sigma_n + T(1-\sigma_n), \tau_n + T(1-\tau_n)) \cdot c(\sigma)^{-1}.\end{aligned}$$

(The element $c(\sigma)$ is obviously defined, since $\sigma(0) = \sigma(1) = 1$.) It is easy to verify that $s_{T-1}(v) = 0$ and $s_T(v) = u$; to complete the proof it suffices to make use of (4.3).

(5.2) THEOREM. *If* D/F *is a central simple algebra of squarefree degree, then* $\bar{K}_2(D) = 0$, *i.e., the natural map*

$$\prod_{L \text{ splits } D} K_2(L) \to K_2(D)$$

is surjective.

PROOF. We first suppose that the degree p of the degree D is a prime. Let F'/F be a maximal extension of degree prime to p. Then, by (1.4), $\bar{K}_2(D) \hookrightarrow \bar{K}_2(D_{F'})$, and it follows from (5.1) that $\bar{K}_2(D_{F'}) = 0$.

In the general case, the degree of D is $p_1 p_2 \cdots p_k$. For any i we can find an extension F_i' of F of degree prime to p_i such that $D_{F_i'}$ is a matrix algebra over a division algebra of degree p_i. By what was proved above, $\bar{K}_2(D_{F_i'}) = 0$; consequently, the group $\bar{K}_2(D)$ has period $[F_i': F]$. Since the greatest common divisor of the numbers $p_1 p_2 \cdots p_k$ and $[F_i': F]$ is 1 and $p_1 p_2 \cdots p_k \cdot K_2(D) = 0$ by (1.4), it follows that $\bar{K}_2(D) = 0$.

(5.3) REMARK. Alperin and Dennis [22] proved a stronger result for quaternion algebras: the map

$$\prod_{\substack{L \text{ a maximal} \\ \text{subfield in } D}} K_2(L) \to K_2(D)$$

is surjective. An analogous result holds for algebras of degree p, if the field F does not have extensions of degree prime to p (this follows from (5.1)). We do not know whether the corresponding assertion is true in the general case.

§6. The reduced norm for K_2

Let D/F be a central simple algebra of squarefree degree. By the reduced norm we mean the homomorphism Nrd: $K_2(D) \to K_2(F)$ such that, for every splitting field L/F of the algebra D ($[L: F] < \infty$), the diagram

$$\begin{array}{ccc} K_2(E) & \overset{\theta_E}{\to} & K_2(D) \\ {\scriptstyle N_{E/F}}\searrow & & \swarrow{\scriptstyle \mathrm{Nrd}} \\ & K_2(F) & \end{array}$$

commutes.

From (5.2) it follows that this property determines the map Nrd uniquely. The existence of the reduced norm will be proved in §7. In this section we shall study some properties of the map Nrd. In what follows we shall only consider algebras of squarefree degree.

(6.1.1) If $D = M_n(T)$, where T is a division algebra, then the map Nrd_D exists if and only if Nrd_T exists, and in this case the diagram

$$\begin{array}{ccc} K_2(D) & \cong & K_2(T) \\ \mathrm{Nrd}\searrow & & \swarrow \mathrm{Nrd}_T \\ & K_2(F) & \end{array}$$

commutes.

(6.1.2) Suppose that F'/F is an arbitrary extension, and the maps Nrd_D and $\mathrm{Nrd}_{D_{F'}}$ exist. Then the following diagram is commutative:

$$\begin{array}{ccc} K_2(D) & \to & K_2(D_{F'}) \\ \downarrow \mathrm{Nrd}_D & & \downarrow \mathrm{Nrd}_{D_{F'}} \\ K_2(F) & \to & K_2(F') \end{array}$$

(6.13) If the extension F'/F is finite, then the diagram

$$\begin{array}{ccc} K_2(D_{F'}) & \overset{N_{D_{F'}/D}}{\to} & K_2(D) \\ \downarrow \mathrm{Nrd}_{D_{F'}} & & \downarrow \mathrm{Nrd}_D \\ K_2(F') & \overset{N_{F'/F}}{\to} & K_2(F) \end{array}$$

commutes.

If E is a splitting field (not necessarily finite) of D, then there exists a canonical map $p_{D/E}$: $K_2(D) \to K_2(E)$, which is defined as the composition $K_2(D) \to K_2(D_E) = K_2(E)$.

(6.1.4) If E/F is a splitting field for D, then the following diagram commutes:

$$\begin{array}{ccc} & K_2(D) & \\ \mathrm{Nrd}\swarrow & & \searrow p_{D/E} \\ K_2(F) & \to & K_2(E) \end{array}$$

(6.1.5) $N_{D/F} = n \cdot \mathrm{Nrd}$, where n is the degree of the algebra D.

All of these facts follow easily from the definition of the reduced norm and the properties of the map θ_L (see §1). We now discuss some sufficient conditions for existence of the reduced norm.

(6.2.1) PROPOSITION. *Suppose that there exists a splitting field E/F of D such that the homomorphism $K_2(F) \to K_2(E)$ is injective. Then the map* Nrd_D *exists.*

PROOF. If L/F is a splitting field for D of finite degree, then the composition $p_{D/E} \cdot \theta_L$ coincides with the map $K_2(L) \xrightarrow{N_{L/F}} K_2(F) \to K_2(E)$, and hence $p(\operatorname{im}\theta) \subset \operatorname{im}(K_2(F) \to K_2(E))$. Using (5.2), we obtain $p(K_2(D)) \in \operatorname{im}(K_2(F) \to K_2(E))$, and we can define the map Nrd using the homomorphism p.

(6.2.2) REMARK. If E is a splitting field for D, then, as shown above,

$$p(K_2(D)) \subset \operatorname{im}(K_2(F) \to K_2(E)).$$

(6.2.3) Suppose that there exist extensions $F_1, \ldots, F_k/F$ such that

a) the maps $\mathrm{Nrd}_{D_{F_i}}$ exist, and

b) the greatest common divisor of the numbers $[F_1: F], \ldots, [F_k: F]$ is 1.

Then the map Nrd_D also exists.

PROOF. We choose integers $m_1, \ldots, m_k$ such that $\sum_1^k m_i \cdot [F_i: F] = 1$. Then the map

$$K_2(D) \to \prod K_2(D_{F_i}) \xrightarrow{\prod \mathrm{Nrd}_{D F_i}} \prod K_2(F_i) \xrightarrow{\prod N_{F_i/F}} \prod K_2(F) \xrightarrow{(m_1, \ldots, m_k)} K_2(F)$$

satisfies all of the conditions in the definition of Nrd.

(6.2.4) COROLLARY. *If the map* Nrd *exists for algebras of prime degree, then it also exists for algebras of squarefree degree.*

(6.3) *Let Λ be an Azumaya algebra of squarefree degree over a "good" regular local ring O, and suppose that the map* Nrd_{Λ_F} *exists. Then* $\mathrm{Nrd}(K_2(\Lambda)) \subset K_2(O)$.

PROOF. Since $K_2(O) = \bigcap_{\mathrm{ht}\,\mu = 1} K_2(O_\mu)$, we may assume that O is a discrete valuation ring. Because the sequence $0 \to K_2(O) \to K_2(F) \xrightarrow{\partial} \mathrm{k}^* \to 0$ is exact, it suffices for us to show that $\partial(\mathrm{Nrd}(K_2(\Lambda)) = 1$. Let $X/\operatorname{Spec} O$ be the Severi-Brauer scheme corresponding to the algebra Λ (see [15]), and let $X_0/\operatorname{Spec} k$ be its closed fiber. Finally, let x_0 be the generic point of X_0. We have the natural homomorphisms $K_2(\Lambda) \to K_2(X)$ and $K_2(\Lambda_F) \to K_2(X_F)$ [20], and the composition $K_2(\Lambda_F) \to K_2(X_F) \to K_2(F(X))$ coincides with $p_{\Lambda_F/F(X)}$. By (6.14) and the commutativity of the diagram

$$\begin{array}{ccccc} K_2(\Lambda) & \to & K_2(X) & \to & K_2(O_{X,x_0}) \\ \downarrow & & \downarrow & & \downarrow \\ K_2(\Lambda_F) & \to & K_2(X_F) & \to & K_2(F(X)) \end{array}$$

we conclude that for any element $u \in K_2(\Lambda)$ we have $\mathrm{Nrd}(u)_{F(X)} \in K_2(O_{X,x_0})$, and hence $\partial_{x_0}(\mathrm{Nrd}(u)_{F(X)}) = 1$.

On the other hand, $\partial_{x_0}(\mathrm{Nrd}(u)_{F(X)}) = \partial(\mathrm{Nrd}(u))_{k(x_0)}$, since O_{X,x_0} is unramified over O.

(6.4) *Let Λ be an Azumaya algebra of squarefree degree over a discrete valuation ring O, and suppose that the degree of Λ is not divisible by* char k. *Further suppose that the map* $\mathrm{Nrd}_{\bar{\Lambda}}$ *exists. If L/F is a splitting field for the algebra Λ_F, then the diagram*

$$\begin{array}{ccccc} K_2(L) & \xrightarrow{N_{L/F}} & K_2(F) & \xrightarrow{s_\pi} & K_2(k) \\ \downarrow \theta_L & & \downarrow & & \| \\ K_2(\Lambda_F) & \xrightarrow{s_\pi} & K_2(\bar{\Lambda}) & \xrightarrow{\mathrm{Nrd}_{\bar{\Lambda}}} & K_2(k) \end{array}$$

commutes.

PROOF. We replace the field L by its maximal separable subextension (which also splits Λ_F); hence, we may suppose that the extension L/F is separable. Then, replacing O by $\hat{O}$, F by $\hat{F} = F \otimes_O \hat{O}$, and L by $\hat{L} = L \otimes_O \hat{O}$, we may assume that O is a complete ring. Finally, replacing L by its maximal unramified subextension, we suppose that the extension L/F is unramified. In this case, $s_\pi \theta_L = \theta_L^- s_\pi$ (2.2.5) and $\mathrm{Nrd}_\Lambda^- s_\pi \theta_L = \mathrm{Nrd}_\Lambda^- \theta_L^- s_\pi = N_{L/k}^- s_\pi = s_\pi N_{L/F}$.

(6.5) COROLLARY. *If the map* $\mathrm{Nrd}\,\Lambda_F$ *also exists, then* $\mathrm{Nrd}_\Lambda^- s_\pi = s_\pi \mathrm{Nrd}_{\Lambda_F}$.

§7. Existence of the reduced norm

(7.1) PROPOSITION. *Suppose that* Λ *is an Azumaya algebra of prime degree* p *over a discrete valuation ring* O, *and* $p \neq \mathrm{char}\, k$. *Suppose that the reduced norm maps exist for the algebras* $O = \Lambda_F$ *and* $D(T) = D \otimes_F F(T)$. *Let* $x_i, y_i, x_i', y_i' \in \Lambda^*$, *and* $\prod[x_i, y_i] = 1 = \prod[x_i', y_i']$, $\bar{x}_i = \bar{x}_i'$, $\bar{y}_i = \bar{y}_i'$. *Set*

$$u = c(x_1, y_1) \cdot \cdots \cdot c(x_n, y_n) \in K_2(\Lambda) \subset K_2(D),$$
$$u' = c(x_1', y_1') \cdot \cdots \cdot c(x_n', y_n').$$

Then $s_\pi(\mathrm{Nrd}\, u) = s_\pi(\mathrm{Nrd}\, u')$.

PROOF. If the map $\mathrm{Nrd}_{\bar{\Lambda}}^-$ exists, then the assertion follows from (6.5). In particular, the proposition is proved if $\bar{\Lambda}$ is not a division algebra. Now suppose that $\bar{\Lambda}$ is a division algebra; we set $x_i' = x_i + \pi x_i''$ and $y_i' = y_i + \pi y_i''$, and we choose polynomial mappings as in (3.3). We consider the following elements:

$$x = \prod [x_i + \pi T x_i'', y_i + \pi T y_i''] \in \mathrm{SL}_1(D(T)),$$
$$w = c(x_i + \pi T x_1'', y_1 + \pi T y_1'') \cdot \cdots \cdot c(x_n + \pi T x_n'', y_n + \pi T y_n'')$$
$$\cdot c(x)^{-1} u^{-1} \in K_2(D(T)),$$
$$v = s_\pi \mathrm{Nrd}\, w \in K_2(k(T)).$$

We let O_0 (respectively, O_1) denote the local ring $O[T]_{(\pi, T)}$ (respectively, $O[T]_{(\pi, T-1)}$), and we set $\Lambda_0 = \Lambda \otimes_O O_0$ and $\Lambda_1 = \Lambda \otimes_O O_1$. From the formula for w we immediately find that $w \in K_2(\Lambda_0) \cap K_2(\Lambda_1)$.

We have the following equalities:

$$s_\pi(\mathrm{Nrd}(u' \cdot u^{-1})) = s_\pi(\mathrm{Nrd}\, s_{T-1}(w)) = s_\pi s_{T-1} \mathrm{Nrd}\, w = s_{T-1}(v)$$

(since $\mathrm{Nrd}\, w \in K_2(O_0) \cap K_2(O_1)$). On the other hand,

$$s_T(v) = s_T s_\pi \mathrm{Nrd}\, w = s_\pi \mathrm{Nrd}\, s_T w = 0$$

(since $s_T w = 0$).

We note that our constructions are preserved under etale extensions of the ring O. It is also easy to see that the existence of the reduced norm for $\bar{\Lambda}(T)$ implies that v is trivial.

We choose an etale extension O'/O such that $\Lambda_{O'}$ is trivial. Then the field k' splits the algebra $\bar{\Lambda}$, and F' splits Λ_F. According to the preceding remark, $v_{k'(T)} = 0$. This means that all of the local invariants of v are trivial; hence, $v \in K_2(k)$, and $v = s_T(v) = 0$.

(7.2) COROLLARY. *Under the conditions of* (7.1), *the map* Nrd_Λ^- *exists.*

PROOF. We may assume that $\bar{\Lambda}$ is a division algebra. Let $\tilde{x}_1, \tilde{y}_1, \ldots, \tilde{x}_n, \tilde{y}_n \in \bar{\Lambda}^*$ and $\prod[\tilde{x}_i, \tilde{y}_i] = 1$. We choose elements $x_i, y_i \in \Lambda^*$ such that $\bar{x}_i = \tilde{x}_i$ and $\bar{y}_i = \tilde{y}_i$, and we set

$x = \prod[x_i, y_i]$ and

$$N(x_1, y_1, \ldots, x_n, y_n) = s_\pi\big(\mathrm{Nrd}\big(c(x_1, y_1) \cdot \cdots \cdot c(x_n, y_n) \cdot c(x)^{-1}\big)\big).$$

According to (7.1), the function N is correctly defined, and it is not hard to see, using (7.1) and Rehmann's representation [21], that N gives a homomorphism $N\colon K_2(\overline{\Lambda}) \to K_2(k)$. In order to prove that N is the reduced norm map, we need the following fact.

(7.2.1) LEMMA. *If k'/k is a separable splitting field for the algebra $\overline{\Lambda}$, then there exists an unramified splitting field F'/F for Λ_F such that $\overline{F'} = k'$.*

(7.2.2) The homomorphism N that was constructed above is uniquely determined by the formula $Ns_\pi = s_\pi \mathrm{Nrd}$. If k' is a separable splitting field for $\overline{\Lambda}$, and F' is the field in (7.2.1), then we have the following commutative diagram:

$$\begin{array}{ccccc} K_2(F') & \overset{\theta_{F'}}{\to} & K_2(\Lambda_F) & \overset{\mathrm{Nrd}}{\to} & K_2(F) \\ \downarrow s_\pi & & \downarrow s_\pi & & \downarrow s_\pi \\ K_2(k') & \overset{\theta_{k'}}{\to} & K_2(\overline{\Lambda}) & \overset{N}{\to} & K_2(k) \end{array}$$

Hence, $N\theta_{k'}s_\pi = s_\pi \mathrm{Nrd}\,\theta_{F'} = s_\pi N_{F'/F} = N_{k'/k}s_\pi$, and so $N\theta_{k'} = N_{k'/k}$.

(7.3) THEOREM. *If D/F is a central simple algebra of squarefree degree prime to* char F, *then the map* Nrd_D *exists.*

PROOF. According to the remark in §6, it suffices to consider the case when the degree of D is a prime p, and the field F does not have extensions of degree prime to p (in particular, $F \supset \mu_p$). We write $D = A_{\xi,F}(a, b)$, and we consider the universal algebra $\tilde{D} = A_{\xi,F(B)}(a, B)$. According to (7.2), it is sufficient to prove that the map Nrd $\tilde{D}$ exists, and for this it suffices to show that $K_2(F(B)) \hookrightarrow K_2(F(B)(X))$, where X is the Severi-Brauer variety corresponding to $\tilde{D}$. If $u \in \ker(K_2(F(B)) \to K_2(F(B)(X)))$ and F'/F is an extension such that the field $F'(B)$ splits the algebra $\tilde{D}$, then $u \in \ker(K_2(F(B)) \to K_2(F'(B)))$, i.e. $u \in K_2(F)$. On the other hand, it is not hard to see that the extension $F(B)(X)/F$ is purely transcendental, and hence $u = 0$.

§8. The K-cohomology of Severi-Brauer varieties

In this section we shall use more or less well-known facts about K-cohomology, characteristic classes, and the Riemann-Roch formula for higher K-groups (see [7], [14], [17], [20] and [25]).

(8.1) **K-cohomology.** If X is an arbitrary scheme, we let $\mathcal{K}_q(\mathcal{O}_X)$ or simply $\mathcal{K}_q$ denote the sheaf on X (in the Zariski topology) associated to the presheaf $U \mapsto K_q(\Gamma(U, \mathcal{O}_X))$. Clearly, for any point $x \in X$ the fiber $(\mathcal{K}_q)_x$ is equal to $K_q(\mathcal{O}_{X,x})$. If $f\colon X \to Y$ is an arbitrary morphism of schemes, we have the sheaf homomorphism $\mathcal{K}_q(\mathcal{O}_Y) \to f^*(\mathcal{K}_q(\mathcal{O}_X))$, which induces a homomorphism of cohomology groups $f^*\colon H^p(Y, \mathcal{K}_q) \to H^p(X, \mathcal{K}_q)$; in this way $H^p(-, \mathcal{K}_q)$ becomes a contravariant functor from the category of schemes to the category of abelian groups.

(8.1.1) *The canonical resolution* [20]. For any smooth variety X over a field F, the sheaf $\mathcal{K}_q(\mathcal{O}_X)$ has a canonical resolution:

$$0 \to \mathcal{K}_q \to \prod_{\mathrm{codim}\, x = 0} (i_x)_*\big(K_q(F(x))\big) \overset{\partial}{\to} \prod_{\mathrm{codim}\, x = 1} (i_x)_*\big(K_{q-1}(F(x))\big) \to \cdots.$$

Hence, the $H^p(X, \mathcal{K}_q)$ coincide with the homology groups of the following complex:

$$\prod_{\operatorname{codim} x=0} K_q(F(x)) \xrightarrow{\partial} \prod_{\operatorname{codim} x=1} K_{q-1}(F(x)) \xrightarrow{\partial} \prod_{\operatorname{codim} x=2} K_{q-2}(F(x)) \xrightarrow{\partial} \cdots.$$

In particular, $H^p(X, \mathcal{K}_q) = 0$ for $p > q$ and $H^p(X, \mathcal{K}_q) = \mathrm{Ch}^p(X)$.

(8.1.2) *The multiplicative structure.* For any smooth variety X, multiplication in K-theory gives a pairing of sheaves

$$\mathcal{K}_p(\mathcal{O}_X) \times \mathcal{K}_n(\mathcal{O}_X) \to \mathcal{K}_{p+n}(\mathcal{O}_X)$$

and consequently a pairing of cohomology groups

$$H^p(X, \mathcal{K}_q) \times H^m(X, \mathcal{K}_n) \to H^{p+m}(X, \mathcal{K}_{q+n}).$$

This pairing determines a bigraded commutative associative ring structure on the group $\Pi_{p,q\geqslant 0} H^p(X, \mathcal{K}_q)$.

If $f\colon X \to Y$ is a morphism of smooth varieties, then $f^*\colon H^*(Y, \mathcal{K}_*) \to H^*(X, \mathcal{K}_*)$ is a ring homomorphism.

(8.1.3) *Covariant properties.* Let $X \to Y$ be a proper morphism of irreducible smooth varieties. We have the homomorphism of complexes (where $d = \dim Y - \dim X$)

$$\begin{array}{ccccccc}
\prod\limits_{\operatorname{codim} x=0} K_q(F(x)) & \to & \prod\limits_{\operatorname{codim} x=1} K_{q-1}(F(x)) & \xrightarrow{\partial} & \cdots \\
\downarrow & & \downarrow & & \\
\prod\limits_{\operatorname{codim} y=d} K_q(F(y)) & \to & \prod\limits_{\operatorname{codim} y=d+1} K_{q-1}(F(x)) & \xrightarrow{\partial} & \cdots
\end{array}$$

which takes the elements of $K_q(F(x))$ to $K_q(F(f(x))$ if $\dim \bar{x} = \dim \overline{f(x)}$, and takes them to zero otherwise.

This homomorphism of complexes determines a homomorphism of K-cohomology groups

$$f_*\colon H^p(X, \mathcal{K}_q) \to H^{p+d}(Y, \mathcal{K}_{q+d}).$$

The map f_* satisfies the projection formula; in particular, if f is a finite morphism of degree n, then $f_* f^* = n$.

(8.1.4) *Excision.* Let $i\colon Z \hookrightarrow X$ be a closed immersion of codimension n, and let $j\colon X - Z = U \hookrightarrow X$. Then we have the exact sequence (we are assuming that the variety Z is smooth and equidimensional)

$$\cdots \xrightarrow{\partial} H^{i-n}(Z, \mathcal{K}_{j-n}) \xrightarrow{i_*} H^i(X, \mathcal{K}_j) \xrightarrow{j^*} H^i(U, \mathcal{K}_j) \xrightarrow{\partial} H^{i-n+1}(Z, \mathcal{K}_{j-n+1}) \to \cdots$$

For arbitrary Z, the map $j^*\colon H^i(X, \mathcal{K}_j) \to H^i(U, \mathcal{K}_j)$ is injective for $i < n$, and is bijective if $i < n - 1$ (see [24] and [25]).

(8.2) **The Gersten spectral sequence.** If Z is a smooth variety, there exists a canonical spectral sequence of cohomological type, called the Gersten spectral sequence:

$$\mathcal{E}(X)\colon E_2^{p,q} = H^p(X, \mathcal{K}_{-q}) \Rightarrow K_{-p-q}(X)$$

(see [20]). The associated filtration $K_*(X)$ coincides with the filtration by the dimension of the support.

(8.2.1) *Functorial properties.* A flat morphism $f\colon X \to Y$ of smooth varieties determines a homomorphism of spectral sequences $f^*\colon \mathcal{E}(Y) \to \mathcal{E}(X)$.

A finite morphism $f\colon X \to Y$ of smooth varieties determines a homomorphism $f_*\colon \mathfrak{S}(X) \to \mathfrak{S}(Y)$.

If the morphism f is flat and finite, then the composition $f_* f^*\colon \mathfrak{S}(Y) \to \mathfrak{S}(X)$ coincides with multiplication by $\deg f$.

(8.3) **K-theory and K-cohomology of projective space.** The K-theory of projective space $X = \mathbf{P}_F^r$ was computed by Quillen in [20]. The exact functors

$$F\text{-mod} \to \mathcal{O}_X\text{-mod}, \qquad V \to V \otimes_F \mathcal{O}_X(-i)$$

determine homomorphisms $K_q(F) \to K_q(X)$, and their direct sum $\prod_{i=0}^{r} K_q(F) \to K_q(X)$ is an isomorphism.

(8.3.1) The K-cohomology of projective space was computed in [24] and [25]. The result can be stated as follows: the canonical homomorphism induced by multiplication

$$\mathrm{Ch}^*(X) \otimes_Z K_*(F) \to H^*\big(X, \mathcal{K}_*\big),$$

is an isomorphism. In other words, $H^p(X, \mathcal{K}_q) \cong K_{q-p}(F)$, where this isomorphism is induced by the maps

$$\begin{array}{c} K_{q-p}(F) \\ \downarrow \\ K_{q-p}(F(x_0)) \\ \downarrow \\ \displaystyle\prod_{\operatorname{codim} x = p-1} K_{q-p+1}(F(x)) \to \prod_{\operatorname{codim} x = p} K_{q-p}(F(x)) \to \prod_{\operatorname{codim} x = p+1} K_{q-p-1}(F(x)) \end{array}$$

where x_0 is the generic point of a linear subvariety of codimension p.

(8.3.2) It follows from (8.3.1) that the Gersten spectral sequence $\mathfrak{S}(X)$ degenerates starting at E_2. Moreover, the filtration by the dimension of the support on $K_*(X)$ is given by the relation $K_*(X)^i = (\gamma - 1)^i \cdot K_*(X)$, where $\gamma = [\mathcal{O}_X(-1)] \in K_0(X)$. In other words, if we identify $K_p(X)$ with $K_p(F) + \gamma K_p(F) + \cdots + \gamma^r K_p(F)$, as in (8.3), then

$$K_p(X)^i = (\gamma - 1)^i K_p(F) + (\gamma - 1)^{i+1} K_p(F) + \cdots + (\gamma - 1)^r K_p(F).$$

(8.4) **Characteristic classes [7], [14].** There exist canonical maps

$$c_{j,i}\colon K_j(X) \to H^{i-j}(X, \mathcal{K}_i),$$

which have the following properties:

a) Functoriality.

b) The maps $c_{0,i}\colon K_0(X) \to H^i(X, \mathcal{K}_i) = \mathrm{Ch}^i(X)$ coincide with the usual Grothendieck characteristic classes.

c) The $c_{j,i}$ are group homomorphisms for $j > 0$.

d) If $\alpha \in K_p(X)$ and $\beta \in K_q(X)$, where $p, q > 0$, then

$$c_{p+q,i}(\alpha\beta) = -\sum_{r+s=i} \frac{(i-1)!}{(r-1)!\,(s-1)!} c_{p,r}(\alpha) c_{q,s}(\beta).$$

e) $c_{1,1}\colon K_1(X) \to H^0(X, \mathcal{K}_1)$ coincides with the natural map induced by the homomorphism $K_1(X) \to K_1(F(X))$.

(8.4.1) If F is an arbitrary field and $X = \operatorname{Spec} F$, then it follows from d), e) and Matsumoto's theorem that the map $c_{2,2}\colon K_2(F) \to H^0(X, \mathcal{K}_2) = K_2(F)$ coincides with multiplication by -1. This fact, along with the functoriality of characteristic classes,

implies that, if X is an arbitrary smooth variety, the map $c_{2,2}\colon K_2(X) \to H^0(X, \mathcal{K}_2)$ is induced up to a sign by the homomorphism $K_2(F) \to K_2(F(X))$.

The product formula d) shows that, for an arbitrary field F, the restriction of the map $c_{n,n}\colon K_n(F) \to H^0(X, \mathcal{K}_n)$ to the subgroup of decomposable elements (i.e, to the image of the homomorphism $K_1(F) \times \cdots \times K_1(F) \xrightarrow{\text{mult}} K_n(F)$) coincides with multiplication by $(-1)^{n-1}(n-1)!$ It has been conjectured (see, for example, [7]) that the map $c_{n,n}\colon K_n(F) \to K_n(F)$ coincides with multiplication by $(-1)^{n-1}(n-1)!$ (at least, modulo torsion); however, examples have recently been constructed which disprove this conjecture.

(8.5) **The Riemann-Roch formula** [7], [14]. We shall only consider the following special case.

Let $i\colon Y \hookrightarrow X$ be a closed immersion of irreducible smooth varieties. Suppose that the conormal sheaf of Y in X is trivial, and let $d = \operatorname{codim}_X Y$. If $y \in K_r(Y)$, then

$$c_{r,q}(i_* y) = (-1)^d \frac{(q-1)!}{(q-d-1)!} i_*(c_{r,q-d}(y)).$$

(8.6) **Degeneration of the Gersten spectral sequence** [7].

(8.6.1) LEMMA. *The map $c_{j,i}\colon K_j(X) \to H^{i-j}(X, \mathcal{K}_i)$ is trivial on the group $K_j(X)^{i-j+1}$. If $j = 0$, then the induced map $K_0(X)^i \to H^i(X, \mathcal{K}_i) = \mathrm{Ch}^i(X)$ is a group homomorphism.*

The first assertion follows from the excision theorem and functoriality of characteristic classes; the second assertion is well known [16].

(8.6.2) Suppose that F is a perfect field, and X is an irreducible smooth variety of dimension n. In what follows we consider the spectral sequence $\mathcal{E}(X) \otimes \mathbf{Z}[1/n!]$.

PROPOSITION. *All of the differentials which enter and leave $E^{p,q}$ are trivial if $p + q = 0$, $p + q = 1$ or $p + q = 2$, and $p < n$. Consequently, in these cases there are natural surjective maps*

$$H^p(X, \mathcal{K}_{-q}) \otimes \mathbf{Z}[1/n!] = E_2^{p,q} \to E_\infty^{p,q} = K_{-p-q}(X)^{p/p+1} \otimes \mathbf{Z}[1/n!].$$

The composition of this map with the homomorphism

$$c_{-p-q,-q}\colon K_{-p-q}(X)^{p/p+1} \otimes \mathbf{Z}[1/n!] \to H^p(X, \mathcal{K}_q) \otimes \mathbf{Z}[1/n!]$$

coincides with multiplication by $(-1)^{-q-1}(-q-1)!$

PROOF. We shall use induction on $s = -p - q$. Since all of the differentials leaving from $E^{p,-s-p}$ go to $E^{i,j}$, where $-i - j = s - 1$, it follows by the induction assumption that they are all zero; consequently, we have a surjective homomorphism

$$H^p(X, \mathcal{K}_{s+p}) \otimes \mathbf{Z}[1/n!] = E_2^{p,-s-p} \to E_\infty^{p,-s-p} = K_s(X)^{p/p+1} \otimes \mathbf{Z}[1/n!].$$

Since all of the differentials leaving from $E^{p,-s-p}$ are trivial, it follows that the natural homomorphism

$$\left(\coprod_{\operatorname{codim} Y = p} K_s'(Y) \right) \otimes \mathbf{Z}\left[\frac{1}{n!}\right] \to H^p(X, \mathcal{K}_{s+p}) \otimes \mathbf{Z}\left[\frac{1}{n!}\right]$$

is surjective (this can be obtained immediately from the construction of the spectral sequence [20]). Thus, in order to prove the last assertion of the proposition, it is sufficient to show that the composition

$$K'_s(Y) \to H^p(X, \mathcal{K}_{s+p}) \otimes \mathbf{Z}[1/n!] \to K_s(X)^{p/p+1} \otimes \mathbf{Z}[1/n!]$$
$$\overset{c_{s,s+p}}{\to} H^p(X, \mathcal{K}_{s+p}) \otimes \mathbf{Z}[1/n!]$$

coincides with the t-fold multiple of the natural map, where $t = (-1)^{q-1}(-q-1)!$. The composition of the first two maps is obviously induced by the immersion $Y \hookrightarrow X$. Using excision, we may assume that Y is a smooth irreducible variety, and the conormal sheaf of Y is trivial. In this case $K'_s(Y) = K_s(Y)$, and we can apply the Riemann-Roch formula, which shows that, for any $y \in K_s(Y)$, the element

$$c_{s,-q}(i_*(y)) = (-1)^p \frac{(-q-1)!}{(s-1)!} i_*(c_{s,s}(y))$$

coincides with the t-multiple of the image of y in the group $H^p(X, \mathcal{K}_{s+p})$, where $t = (-1)^{-q-1}(-q-1)!$ (see (8.4.1)). Thus, we find that the natural homomorphism $E_2^{p,q} \to E_\infty^{p,q}$ is an isomorphism, and so all of the differentials coming into $E^{p,q}$ are zero.

(8.7) **K-cohomology of Severi-Brauer varieties.** Let D be a central simple algebra of prime degree p over a field F ($p \neq \operatorname{char} F$), and let X/F be the corresponding Severi-Brauer variety.

(8.7.1) PROPOSITION. *All of the differentials in $\mathcal{E}(X)$ which enter or leave $E^{i,j}$ for $i+j = 0, -1, -2$ (in the last case we also assume that $i < p-1 = \dim X$) are trivial.*

PROOF. Let L/F be a splitting field for the algebra D of degree p. Since the morphism $\pi\colon X_L \to X$ is flat and finite, it follows that we have natural maps of spectral sequences $\mathcal{E}(X) \overset{\pi^*}{\to} \mathcal{E}(X_L) \overset{\pi_*}{\to} \mathcal{E}(X)$, and their composition coincides with multiplication by p. Since $X_L = \mathbf{P}_L^{p-1}$, all of the differentials in $\mathcal{E}(X_L)$ are zero; consequently, all of the differentials in $\mathcal{E}(X)$ are killed by multiplication by p. On the other hand, according to (8.6), all of the differentials we are considering become zero under localization with respect to powers of $(p-1)!$ Consequently, these differentials are zero.

(8.7.2) COROLLARY. *Let $\bar{F}$ be the algebraic closure of the field F, and let $\bar{X} = X_{\bar{F}}$. Then the natural projection $\bar{X} \to X$ induces isomorphisms (we are assuming that the algebra D is nontrivial)*:

$$\mathrm{Ch}^i(X) \tilde{\to} p\mathbf{Z} \subset \mathbf{Z} = \mathrm{Ch}^i(\bar{X}) \qquad (1 \leqslant i \leqslant p-1),$$

$$H^i(X, \mathcal{K}_{i+1}) \tilde{\to} N \subset \bar{F}^* = H^i(\bar{X}, \mathcal{K}_{i+1}) \qquad (1 \leqslant i \leqslant p-1),$$

where $N = \operatorname{Nrd} D^*$.

PROOF. According to (8.7.1), we have $H^i(X, \mathcal{K}_j) = K_{j-i}(X)^{i/i+1}$ where $0 \leqslant j-i \leqslant 1$. It follows from [20] that

$$K_s(X) = K_s(F) \oplus K_s(D) \oplus \cdots \oplus K_s(D^{\otimes(p-1)}).$$

In particular,

$$K_0(X) = \underbrace{\mathbf{Z} \oplus \mathbf{Z} \oplus \cdots \oplus \mathbf{Z}}_{p}, \qquad K_1(X) = F^* \oplus \underbrace{N \oplus \cdots \oplus N}_{p-1},$$

where the maps $K_0(X) \to K_0(\bar{X})$ and $K_1(X) \to K_1(\bar{X})$ are given by the correspondence

$$(n_0, \ldots, n_{p-1}) \to (n_0, pn_1, \ldots, pn_{p-1}); \ (a, b_1, \ldots, b_{p-1}) \to (a, b_1, \ldots, b_{p-1})$$

and so are injective.

(8.7.2.1) $K_s(X)^i = K_s(X) \cap K_s(\bar{X})^i$ $(s = 0, 1)$.

PROOF. The inclusions $K_s(X)^i \subset K_s(X) \cap K_s(\bar{X})^i$ are obvious. Furthermore, the group $K_s(X) \cap K_s(\bar{X})^i$ coincides with $N_s \cdot [(\gamma - 1)^i, (\gamma - 1)^{i+1}, \ldots, (\gamma - 1)^{p-1}]$, where $N_0 = p\mathbf{Z}$, $N_1 = N$, $\gamma = [O(-1)] = (0, 1, 0, \ldots, 0)$ and $i > 0$. Thus, it suffices to prove that $p(\gamma - 1)^i \in K_0(X)^i$ and $a(\gamma - 1)^i \in K_1(X)^i$ for any $a \in N$. Let $a = N_{L/F}(t)$, where L/F is a splitting field for the algebra D of degree p. We consider the following diagram:

$$\begin{array}{ccccc} & & X_L \times_X \bar{X} & & \\ & g' \swarrow & & \searrow f' & \\ X_L & & & & \bar{X} \\ & f \searrow & & \swarrow g & \\ & & X & & \end{array}$$

We have the equalities

$$g^* f_* \big((\gamma - 1)^i\big) = (f')_* (g')^* \big((\gamma - 1)^i\big) = p(\gamma - 1)^i,$$

$$g^* f_* \big(t(\gamma - 1)^i\big) = (f')_* (g')^* \big(t \cdot (\gamma - 1)^i\big) = a(\gamma - 1)^i.$$

(8.7.3) COROLLARY. *If F'/F is an arbitrary extension, the natural map $H^1(X, \mathcal{K}_2) \to H^1(X_{F'}, \mathcal{K}_2)$ is injective.*

(8.7.4) REMARK. For any point $x \in X$ of codimension 1, the composition $F^* \to H^1(X, \mathcal{K}_2) \xrightarrow{\sim} N$, in which the first map is obtained using the homomorphism of complexes

$$\begin{array}{ccccc} & & F^* & & \\ & & \downarrow & & \\ & & F(x)^* & & \\ & & \downarrow & & \\ K_2(F(X)) & \to & \prod_{\operatorname{codim} z = 1} F(z)^* & \to & \prod_{\operatorname{codim} z = 2} \mathbf{Z} \end{array}$$

coincides with raising to the $(\deg \bar{x})$-power; in particular, it is trivial on the subgroup $\mu_p \cap F^*$.

(8.7.5) COROLLARY. *Under the conditions of* (8.7.1), *the natural map $K_2(F) \to K_2(F(X))$ induces an epimorphism $K_2(F) \to H^0(X, \mathcal{K}_2)$.*

PROOF. Since all of the differentials which leave from $E^{0,-2}$ are equal to zero, we find that the map $K_2(F) \oplus K_2(D) \oplus \cdots \oplus K_2(D^{\otimes(p-1)}) = K_2(X) \to H^0(X, \mathcal{K}_2)$ is surjective. Hence, it suffices to prove that, for any $i = 1, \ldots, p-1$, the image of the group $K_2(D^{\otimes i})$ in $K_2(F(X))$ is contained in the image of $K_2(F)$; but this follows from (6.2.2).

(8.7.6) PROPOSITION. *Under the conditions of* (8.7.1), *there is an isomorphism $K_2(F) \xrightarrow{\sim} H^0(X, \mathcal{K}_2)$.*

PROOF. According to (8.7.5), it suffices for us to prove that $K_2(F) \hookrightarrow K_2(F(X))$. A standard application of the transfer homomorphism shows that the kernel of the map $K_2(F) \to K_2(F(X))$ is annihilated by multiplication by p; consequently, it suffices to consider the following homomorphism:

$$K_2(F) \otimes \mathbf{Z}[1/(p-1)!] \to H^0(X, \mathcal{K}_2) \otimes \mathbf{Z}[1/(p-1)!].$$

This map coincides up to a sign with the restriction to $K_2(F)$ of the class $c_{2,2}$; in addition, $c_{2,i}|_{K_2(F)} = 0$ for $i > 2$ (this follows from functoriality of the maps $c_{2,i}$ and the fact that $H^s(\operatorname{Spec} F, -) = 0$ for $s > 0$). Thus, instead of considering the map $c_{2,2}|_{K_2(F)}$, we shall consider the restriction to $K_2(F)$ of the total class $c_{2,*}$: $K_2(X) \to \prod_{i=0}^{p-1} H^i(X, \mathcal{K}_{i+2})$. It follows from (8.6.2) that

$$\ker(c_{2,*}) \subset K_2(X)^{p-1} \otimes \mathbf{Z}[1/(p-1)!]$$

and so it suffices to prove the following assertion.

(8.7.6.1) LEMMA. $K_2(F) \cap K_2(X)^{p-1} = 0$.

PROOF. The group $K_2(X)^{p-1}$ coincides with the sum of the images of the maps

$$K_2(F(x)) \to K_2(X) = K_2(F) \oplus K_2(D) \oplus \cdots \oplus K_2(D^{\otimes(p-1)}),$$

where x runs through the set of closed points in X. From the definition of the isomorphism $K_2(X) \cong K_2(F) \oplus K_2(D) \oplus \cdots \oplus K_2(D^{\otimes(p-1)})$ it is not hard to conclude that the map $K_2(F(x)) \to K_2(X) = K_2(F) \oplus K_2(D) \oplus \cdots \oplus K_2(D^{\otimes(p-1)})$ is equal to $(N_{F(x)/F}, (1-p)\theta_{F(x)}, \ldots)$; consequently, every element $y = (y_0, y_1, \ldots) \in K_2(X)^{p-1}$ satisfies the equation $y_0 = \operatorname{Nrd}(y_1)$, i.e. $K_2(F) \cap (K_2(X)^{p-1}) = 0$.

(8.7.7) REMARK. There was a small gap in the proofs of (8.7.1) and (8.7.6): we used (8.6.2), where we had assumed that F is a perfect field. In the general case we need only note that the group $\ker(K_2(F) \to K_2(F(X)))$ is annihilated by multiplication by p, and hence this group maps injectively into the corresponding group for the field $F^{l^{-\infty}}$, where $l = \operatorname{char} F$.

(8.7.8) Suppose that X is a smooth variety over a field F which contains the group μ_n of roots of unity. We have the following flabby resolution of the sheaf ${}_n\mathcal{K}_2$:

$$0 \to {}_n\mathcal{K}_2 \to \prod_{\operatorname{codim} x = 0} (i_x)_*(nK_2(F(x))) \xrightarrow{\partial} \prod_{\operatorname{codim} x = 1} (i_x)_*(\mu_n) \to 0$$

and so the following equalities hold for an irreducible variety X:

a) $H^0(X, \mathcal{K}_2) = \{u \in {}_n K_2(F(X)) : \partial_x(u) = 1 \ \forall x : \operatorname{codim} x = 1\}$.

b) $H^1(X, {}_n\mathcal{K}_2) = (\prod_{\operatorname{codim} x = 1} \mu_n)/\partial({}_n K_2(F(X)))$.

c) $H^i(X, {}_n\mathcal{K}_2) = 0$, if $i \geqslant 2$.

The exact sequences $0 \to {}_n\mathcal{K}_2 \to \mathcal{K}_2 \to n\mathcal{K}_2 \to 0$ and $0 \to n\mathcal{K}_2 \to \mathcal{K}_2 \to \mathcal{K}_2/n\mathcal{K}_2 \to 0$ and the above computations yield the following statements:

d) The map $H^1(X, \mathcal{K}_2) \to H^1(X, n\mathcal{K}_2)$ is surjective,

$$H^i(X, \mathcal{K}_2) \xrightarrow{\sim} H^i(X, n\mathcal{K}_2) \quad \text{for } i \geqslant 2.$$

e) The sequence

$$H^i(X, \mathcal{K}_2) \xrightarrow{n} H^1(X, \mathcal{K}_2) \to H^1(X, \mathcal{K}_2/n\mathcal{K}_2) \to {}_n\mathrm{Ch}^2(X) = {}_n H^2(X, \mathcal{K}_2) \to 0$$

is exact.

f) We have the exact sequences

$$\prod_{\operatorname{codim} x=1} \mu_n \to H^1(X, \mathcal{K}_2) \to H^1(X, n\mathcal{K}_2) \to 0,$$
$$H^0(X, \mathcal{K}_2) \to H^0(X, \mathcal{K}_2/n\mathcal{K}_2) \to H^1(X, n\mathcal{K}_2) \to H^1(X, \mathcal{K}_2).$$

(8.7.9) PROPOSITION. *Under the conditions of* (8.7.2), *we have an isomorphism* $H^1(X, \mathcal{K}_2/p\mathcal{K}_2) \cong N/N^p$ *and an exact sequence*

$$K_2(F)/pK_2(F) \to H^0(X, \mathcal{K}_2/p\mathcal{K}_2) \to N \cap \mu_p \to 0.$$

PROOF. We first suppose that $F \subset \mu_p$. Statement e) implies existence of an isomorphism $N/N^p \tilde{\to} H^1(X, \mathcal{K}_2/p\mathcal{K}_2)$, while f) and (8.7.4) give an isomorphism $H^1(X, \mathcal{K}_2) \tilde{\to} H^1(X, p\mathcal{K}_2)$; consequently,

$$\ker\bigl(H^1(X, p\mathcal{K}_2) \to H^1(X, \mathcal{K}_2)\bigr) \tilde{\to} {}_pH^1(X, \mathcal{K}_2) = {}_pN = N \cap \mu_p.$$

Finally, the second sequence in f) gives us the required assertion.

The general case is obtained from this one by means of the transfer map.

§9. Injectivity of the homomorphism $R_{F,p}$

In this section we shall assume that all of our fields contain the group μ_p of pth roots of unity. Let $\xi \in \mu_p$ be a fixed primitive root. The choice of ξ determines an isomorphism $\mu_p \cong \mu_p^{\otimes 2}$, and after this identification the norm residue homomorphism

$$R_{F,p}\colon K_2(F)/pK_2(F) \to H^2\bigl(F, \mu_p^{\otimes 2}\bigr) = H^2(F, \mu_p) = {}_p\mathrm{Br}(F)$$

is given by the correspondence $\{a, b\} \mapsto A_\xi(a, b)$, where $A_\xi(a, b)$ is a cyclic algebra. We shall let $k_2(F)$ denote the group $K_2(F)/pK_2(F)$.

(9.1) PROPOSITION. *Let X be the Severi-Brauer variety corresponding to a nontrivial algebra $A_\xi(a, b)$. Then the group*

$$\bigl\{u \in K_2(F(X))\colon \partial_x(u) \in \mu_p \text{ for any } x\colon \operatorname{codim} x = 1\bigr\}/K_2(F) + \{\xi, F(X)^*\}$$

is cyclic of order p; it is generated by an element $u_0 \in K_2(F(X))$ such that $pu_0 = \{a, b\}$.

PROOF. Since the algebra $A_{\xi, F(X)}(a, b)$ is trivial, we have $\{a, b\} \in pK_2(F(X))$ ([3], §15). Let $\{a, b\} = pu_0$, where $u_0 \in K_2(F(X))$ obviously gives a nonzero element of our group. Thus, it suffices for us to prove that our group contains no more than p elements. If $\partial_x(u) \in \mu_p$ for any x and $\partial_x(u) = \xi^{n_x}$, where only finitely many of the n_x are nonzero, then we can consider the divisor $\Sigma n_x \cdot x$, which is uniquely defined modulo $p\operatorname{Div} X$. If $\Sigma n_x \cdot x = (f)$, then $u - \{\xi, f\} \in K_2(F)$; consequently, u determines a trivial element of our group. This shows that the correspondence $u \to \Sigma n_x x$ gives an injective homomorphism from our group to the group $\operatorname{Pic} X/p \operatorname{Pic} X = \mathbf{Z}/p\mathbf{Z}$.

(9.2) COROLLARY. *Under the conditions of* (9.1),

$${}_pK_2(F(X)) = {}_pK_2(F) + \{\xi, F(X)^*\}.$$

PROOF. Suppose that $u \in {}_pK_2(F(X))$. Using (9.1), we represent u in the form $u = nu_0 + v + \{\xi, f\}$, where $v \in K_2(F)$ and $f \in F(X)^*$. Then $0 = pu = n\{a, b\} + pv$, and so $n \in p\mathbf{Z}$, $n = pm$. Finally, $u = m\{a, b\} + v + \{\xi, f\}$, where $m\{a, b\} + v \in {}_pK_2(F)$.

(9.3) COROLLARY. *Under the conditions of* (9.1), *there is an exact sequence*

$$0 \to \mathbf{Z}/p\mathbf{Z} \xrightarrow{\{a,\,b\}} k_2(F(X)).$$

PROOF. The kernel of the map $k_2(F) \to k_2(F(X))$ coincides with the group

$$\begin{aligned}\{v \in K_2(F)\colon v \in pK_2(F(X))\}/pK_2(F) &= \{u \in K_2(F(x))\colon pu \in K_2(F)\}/K_2(F) + pK_2(F(x)) \\ &= \{u \in K_2(F(x))\colon \partial_x(u) \in \mu_p\ \forall x\}/K_2(F) + \{\xi, F(X)^*\}\end{aligned}$$

and therefore is a cyclic group of order p.

(9.4) PROPOSITION. *The homomorphism* $R_{F,p}\colon k_2(F) \to {}_p\mathrm{Br}(F)$ *is injective.*

PROOF. Let $x = \Sigma_1^n\{a_i, b_i\} \in \ker R_{F,p}$; we shall prove that $x \in pK_2(F)$ by induction on n. The case $n = 1$ is well known ([3], §15). If $\{a_n, b_n\} \in pK_2(F)$, then $\Sigma_1^{n-1}\{a_i, b_i\} \in \ker R_{F,p}$ and, by the induction assumption, $\Sigma_1^{n-1}\{a_i, b_i\} \in pK_2(F)$, i.e. $\Sigma_1^n\{a_i, b_i\} \in pK_2(F)$.

Now suppose that $\{a_n, b_n\} \notin pK_2(F)$, i.e. $A_\xi(a_n, b_n)$ is a nontrivial cyclic algebra. Let X denote the corresponding Severi-Brauer variety. Since $\{a_n, b_n\} \in pK_2(F(X))$, we find, as above, that $\Sigma_1^n\{a_i, b_i\} \in pK_2(F(X))$ and, by (9.3), $\Sigma_1^n\{a_i, b_i\} \equiv m\{a_n, b_n\}$ (mod $pK_2(F)$) for some m. Because the algebra $A_\xi(a_n, b_n)$ is nontrivial, the relation

$$0 = R_{F,p}\left(\sum_{i=1}^{n}\{a_i, b_i\}\right) = mR_{F,p}(\{a_n, b_n\}) = m\left[A_\xi(a_n, b_n)\right]$$

shows that $m \in p\mathbf{Z}$, i.e., $\Sigma_1^n\{a_i, b_i\} \in pK_2(F)$.

§10. The Brauer group of a field with trivial k_2 group

We keep the notation and assumptions of the previous section.

(10.1) Let E/F be a cyclic extension of degree p. We consider the following properties:

(10.1.1) $\ker(N_{E/F}\colon k_2(E) \to k_2(F)) = k_2(E)^{1-\sigma} + k_2(F)$, where σ is a generator of the group $\mathrm{Gal}(E/F)$.

(10.1.2) $\ker(N_{E/F}\colon K_2(E) \to K_2(F)) = K_2(E)^{1-\sigma}$.

(10.2) LEMMA. (10.1.2) $\Rightarrow$ (10.1.1). *If* ${}_pK_2(F) = \{\xi, F^*\}$, *then* (10.1.1) $\Rightarrow$ (10.1.2).

PROOF. Suppose that $u \in K_2(E)$ and $N_{E/F}(u) = pv$ for some element $v \in K_2(F)$. Then $N_{E/F}(u - v_E) = 0$, and, by (10.1.2), $u - v_E \in K_2(E)^{1-\sigma}$; hence, u (mod $pK_2(F)$) $\in k_2(F) + k_2(E)^{1-\sigma}$.

Now suppose that ${}_pK_2(F) = \{\xi, F^*\}$ and (10.1.1) holds. If $N_{E/F}(u) = 0$, it follows from (10.1.1) that $u \in K_2(E)^{1-\sigma} + K_2(F) + pK_2(E) = K_2(E)^{1-\sigma} + K_2(F)$. If we set $u = w^{1-\sigma} + v_E$, where $v \in K_2(F)$, then $0 = N_{E/F}(u) = pv$, and, by assumption, $v = \{\xi, f\}$. Since $E = F(\sqrt[p]{a})$ for some $a \in F^*$, it follows that $v_E = \{\xi, f\}_E = \{\sqrt[p]{a}, f\}^{1-\sigma}$ and $u = (w + \{\sqrt[p]{a}, f\})^{1-\sigma}$.

(10.3) LEMMA. *If the extension* E/F *has the property* (10.1.2), *so does* $E(T)/F(T)$.

The proof follows immediately from the description of K_2 of a rational function field [19] and Hilbert's Theorem 90.

(10.4) LEMMA. *A cyclic extension* $F(Y)/F(T)$, *where* $T = Y^p$, *has the property* (10.1.2) *if and only if* ${}_pK_2(F) = \{\xi, F^*\}$.

PROOF. From Hilbert's Theorem 90 it follows that

$$\ker\Big(K_2(F(Y)) \xrightarrow{N} K_2(F(T))\Big) = K_2(F(Y))^{1-\sigma} +_p K_2(F).$$

Thus, it suffices to prove that $K_2(F) \cap K_2(F(Y))^{1-\sigma} = \{\xi, F^*\}$. Let $u^{1-\sigma} \in K_2(F)$; the local invariants of the element u are σ-invariant, and, since the extension $F(Y)/F(T)$ ramifies only at the point $T = 0$, it follows that there exists $v \in K_2(F(T))$ such that the element $v_{F(Y)}$ has the same local invariants as u at all places except for $Y = 0$. Consequently, $u = v_{F(Y)} + \{Y, c\} + u_0$, where $c \in F^*$, $u_0 \in K_2(F)$, and hence $u^{1-\sigma} = \{\xi, c\}$.

(10.5) LEMMA. *Suppose that E/F is a cyclic extension of degree p, $b \in F^*$, $\lambda \in E^*$ and $N_{E/F}(\{b, \lambda\}) = 0 \in k_2(F)$. Then $\{b, \lambda\} \in k_2(E)^{1-\sigma} + k_2(F)$.*

PROOF. Let $E = F(\sqrt[p]{a})$; since $\{b, N\lambda\} = N\{b, \lambda\} = 0 \in k_2(F)$, it follows that $b = N_{F(\sqrt[p]{N\lambda})}(t)$ for some $t \in F(\sqrt[p]{N\lambda})$. We write the elements λ and t in the form

$$\lambda = x_0 + x_1\sqrt[p]{a} + \cdots + x_{p-1}\sqrt[p]{a^{p-1}} \quad \text{and} \quad t = y_0 + y_1\sqrt[p]{N\lambda} + \cdots + y_{p-1}\sqrt[p]{(N\lambda)^{p-1}}.$$

Substituting these values in the formula for b, we obtain an equality $b = Q(a, x, y)$, where Q is some universal polynomial. Let k be the minimal subfield of F containing μ_p. Then k is either a global field or a finite field, and in either case ${}_pK_2(k) = \{\xi, k^*\}$ (see [26]). We set

$$\tilde{F} = k(A, X_i, Y_j), \quad \tilde{E} = \tilde{F}\Big(\sqrt[p]{A}\Big), \quad B = Q(A, X, Y), \quad \Lambda = \sum_{i=0}^{p-1} X_i \cdot \sqrt[p]{A^i} \in \tilde{E}.$$

Then $\{B, \Lambda\} \in \ker(N\colon k_2(\tilde{E}) \to k_2(\tilde{F}))$ and, by (10.3) and (10.4), $\{B, \Lambda\} \in K_2(\tilde{E})^{1-\sigma} + K_2(\tilde{F})$. To complete the proof it now suffices to apply the specialization homomorphism.

(10.6) PROPOSITION. *Suppose that the field F does not have extensions of degree prime to p, and E/F is a finite extension. If $k_2(F) = 0$, then $k_2(E) = 0$.*

PROOF. We may assume that E/F is a cyclic extension of degree p. In this case, the group $K_2(E)$ is generated by the symbols $\{b, \lambda\}$, where $b \in F^*$ and $\lambda \in E^*$ (see [11]). Since $k_2(F) = 0$, if we apply (10.5) repeatedly, we see that $k_2(E) = k_2(E)^{1-\sigma} = \cdots = k_2(E)^{(1-\sigma)^{p-1}}$. It remains to note that the map $(1-\sigma)^{p-1}\colon k_2(E) \to k_2(E)$ coincides with the composition $k_2(E) \xrightarrow{N_{E/F}} k_2(F) \to k_2(E)$, and therefore is equal to zero.

(10.7) COROLLARY. *Under the conditions of* (10.6), $\mathrm{Br}(F) = 0$.

PROOF. Suppose that D/F is a nontrivial central simple algebra, and L/F is a splitting field of minimal degree for this algebra. We can find an intermediate field E such that $[L: E] = p$. The algebra D_E is cyclic, and, by (10.6), it is trivial. This is a contradiction.

§11. The fundamental theorem

Everywhere (except in Theorem 11.5) we keep the notation and assumptions of the previous section.

(11.1) LEMMA. *Let D/F be a nontrivial cyclic algebra of degree p, and let X/F be the corresponding Severi-Brauer variety. If $\mu_p \subset N = \mathrm{Nrd}\, D^*$, then there exists a central simple algebra D' of exponent p^2 such that $[D'_{F(X)}] \in \mathrm{im}\, R_{F(X),p}$ and $p[D'] = D$.*

PROOF. It suffices to consider the case when the field F does not have extensions of degree prime to p (one uses the transfer map). Then $D = A_\xi(a, b)$, and we may assume that $\xi \in \operatorname{im} N_{F(\sqrt[p]{a})/F}$. Identifying the groups $\mathbf{Z}/p\mathbf{Z}$ and μ_p, we find that the homomorphism δ in the exact sequence

$$H^1(F, \mathbf{Q}/\mathbf{Z}) \xrightarrow{p} H^1(F, \mathbf{Q}/\mathbf{Z}) \xrightarrow{\delta} H^2(F, \mathbf{Z}/p\mathbf{Z}) = {}_p\mathrm{Br}(F)$$

takes the character χ_c corresponding to the extension $F(\sqrt[p]{c})/F$ to $[A_\xi(c, \xi)] \in {}_p\mathrm{Br}(F)$. Since the algebra $A_\xi(a, \xi)$ is trivial, it follows from the exact sequence that the character χ_a is divisible by p, i.e., the field $F(\sqrt[p]{a})$ is imbedded in a cyclic extension L/F of degree p^2. We consider the cyclic algebra $D' = (L/F, b)$. Since $p[D'] = [A_\xi(a, b)] \neq 0$, the algebra D' has exponent p^2. We set $F' = F(X)$ and $L' = L \otimes_F F'$. Since the algebra $A_{\xi, F'}(a, b)$ is trivial, there exists an element $t \in F'(\sqrt[p]{b})$ such that $a = N_{F'(\sqrt[p]{b})/F'}(t)$. Let σ be a generator of the group $\mathrm{Gal}(F'(\sqrt[p]{b})/F')$. Then

$$\left\{t, \sqrt[p]{b}\right\}^{1+\sigma+\cdots+\sigma^{p-1}} = \left\{a, \sqrt[p]{b}\right\} + \left\{t^{\tilde{N}}, \xi\right\},$$

where $\tilde{N} = \sigma + 2\sigma^2 + \cdots + (p-1)\sigma^{p-1}$. The norm of the element $\{t, \xi\}$ is equal to $\{a, \xi\} = 0 \in k_2(F')$; consequently, by (10.5), $\{t, \xi\} \in k_2(L')^{1-\sigma} + k_2(F')$. Thus, $\{t^{\tilde{N}}, \xi\}$, and hence also the element $\{a, \sqrt[p]{b}\}$, belong to the image of the homomorphism $k_2(F') \to k_2(L')$, i.e., there exists a central simple F'-algebra T such that $[T \otimes_{F'} L'] = [A_{\xi, L'}(a, \sqrt[p]{b})] = [D' \otimes_{F'} L']$ and $[T] \in \operatorname{im} R_{F',p}$. Consequently, the element $[D'_{F'}]$ differs from $[T]$ by the class of some cyclic algebra of degree p, which obviously belongs to the group $\operatorname{im} R_{F',p}$; hence, $[D'_{F'}] \in \operatorname{im} R_{F',p}$.

(11.2) LEMMA. *The group*

$$G = \left\{[D'] \in \mathrm{Br}(F) : \left[D'_{F(x)}\right] \in \operatorname{im} R_{F(x),p}\right\} / \operatorname{im} R_{F,p}$$
$$= \{u \in K_2(F(x)) : \partial_x(u) \in F(x)^{*p} \forall x\} / pK_2(F(X)) + K_2(F)$$

is isomorphic to $N \cap \mu_p$, *where* $N = \mathrm{Nrd}\, D^*$.

PROOF. Let $u \in K_2(F(X))$ and $\partial_x(u) = a_x^p$ for any point x of codimension 1. It is clear that $\{a_x\}_{\mathrm{codim}\, x = 1}$ determines an element $a(u) \in H^1(X, \mathcal{K}_2) = N$, which does not depend on the choice of a_x; and $a(u) = 1$ if and only if $u \in pK_2(F(X)) + K_2(F)$. In addition, $a(u)^p = 1$, and so we obtain an injective homomorphism a: $G \hookrightarrow N \cap \mu_p$. Its image coincides with $N \cap \mu_p$, in view of (11.1).

(11.3) COROLLARY. $\operatorname{coker} R_{F,p} \hookrightarrow \operatorname{coker} R_{F(X),p}$.

(11.4) PROPOSITION. *The homomorphism* $R_{F,p}$: $k_2(F) \to {}_p\mathrm{Br}(F)$ *is surjective.*

PROOF. We define a chain of imbedded fields $F = F_0 \subset F_1 \subset F_2 \subset \cdots$ in the following way: if i is even, then F_{i+1} is the maximal algebraic extension of F_i of degree prime to p; if i is odd, then F_{i+1} is the compositum of the function fields of the Severi-Brauer varieties corresponding to all possible cyclic algebras of degree p over F_i. We set $\tilde{F} = \bigcup_i F_i$. Using (11.3), along with the well-known fact that $\operatorname{coker} R_{F,p} \hookrightarrow \operatorname{coker} R_{E,p}$ if the degree of the extension E/F is not divisible by p, we see that the field $\tilde{F}$ has the following properties:

a) $\tilde{F}$ does not have extensions of degree prime to p.

b) $k_2(\tilde{F}) = 0$.

c) $\operatorname{coker} R_{F,p} \hookrightarrow \operatorname{coker} R_{\tilde{F},p}$.

Then, by (10.7), we have $\operatorname{Br}(\tilde{F}) = 0$, and hence $\operatorname{coker} R_{\tilde{F},p} = 0$.

(11.5) THEOREM. *Let F be an arbitrary field, and let n be a natural number prime to* char F. *Then the map*

$$R_{F,n}\colon K_2(F)/nK_2(F) \to H^2\big(F, \mu_n^{\otimes 2}\big)$$

is an isomorphism.

PROOF. We may obviously assume that $n = p^k$, where p is a prime, and $\mu_p \subset F$ (where one uses the transfer homomorphism). The proof proceeds by induction on k. The case $k = 1$ was considered above. We have the following commutative diagram:

$$\begin{array}{ccccccc} F^*/F^{*p} & \otimes \mu_p & \to & K_2(F)/p^k & \xrightarrow{p} & K_2(F)/p^{k+1} & \to & K_2(F)/p \to 0 \\ \downarrow & & & \downarrow R_{F,p^k} & & \downarrow R_{F,p^{k+1}} & & \downarrow R_{F,p} \\ H^1\big(F, \mu_p\big) & \otimes \mu_p & \to & H^2\big(F, \mu_{p^k}^{\otimes 2}\big) & \to & H^2\big(F, \mu_{p^{k+1}}^{\otimes 2}\big) & \to & H^2\big(F, \mu_p^{\otimes 2}\big) \end{array}$$

Both rows of this diagram are exact (with the possible exception of the $K_2(F)/p^k$ term). A simple diagram chasing shows that the map $R_{F,p^{k+1}}$ is an isomorphism.

§12. The reduced norms

(12.1) THEOREM. *Let $F \supset \mu_p \ni \xi \neq 1$, $D = A_\xi(a, b)$ and $c \in F^*$. Then the following conditions are equivalent*:

a) $c \in N = \operatorname{Nrd} D^*$.

b) $\{a, b, c\} \in pK_3^M(F)$.

c) $\chi_a \chi_b \chi_c = 0 \in H^3(F, \mu_p)$.

PROOF. The implications a) $\Rightarrow$ b) $\Rightarrow$ c) are obvious; we now show that c) $\Rightarrow$ a). We may assume that the algebra D is nontrivial. We let X denote the Severi-Brauer variety corresponding to D. Using Bloch's exact sequence from [13] along with (11.5), we obtain

1) $H^0(X, \mathcal{K}_2/p\mathcal{K}_2) \xrightarrow{\sim} {}_p\operatorname{Br}(X)$, and

2) $H^1(X, \mathcal{K}_2/p\mathcal{K}_2) \hookrightarrow H^3(X, \mu_p)$.

From the results of §8 it follows that $H^1(X, \mathcal{K}_2/p\mathcal{K}_2) \simeq N/N^p$. The Hochschild-Serre spectral sequence gives us the following exact sequence of etale cohomology:

$$0 \to H^2\big(F, \mu_p\big) \to H^2\big(X, \mu_p\big) \to \mathbf{Z}/p\mathbf{Z} \xrightarrow{\chi_a \chi_b \chi_\xi} H^3\big(F, \mu_p\big) \to H^3\big(X, \mu_p\big) \to H^1\big(F, \mu_p\big) = F^*/F^{*p}.$$

It is easy to verify that the composition

$$N/N^p = H^1\big(X, \mathcal{K}_2/p\mathcal{K}_2\big) \to H^3\big(X, \mu_p\big) \to F^*/F^{*p}$$

is induced by the imbedding $N \hookrightarrow F^*$, and the diagram

$$\begin{array}{ccc} & F^* & \xrightarrow{c \mapsto \chi_a \chi_b \chi_c} H^3\big(F, \mu_p\big) \\ \overset{c}{\underset{c^p \bmod N^p}{\downarrow}} & \downarrow & \downarrow \\ & N/N^p & \to H^3\big(X, \mu_p\big) \end{array}$$

is commutative. Therefore, $c^p \in N^p$, i.e. $c = c_0 \cdot \xi^r$, where $c_0 \in N$; consequently, it is sufficient to consider the case $c = \xi$. In this case the map $H^2(F, \mu_p) \to H^2(X, \mu_p)$ is not surjective. From the exact sequence

$$0 \to \mathbf{Z}/p\mathbf{Z} \xrightarrow{\pi^*(\chi_a \chi_b)} H^2(X, \mu_p) \to {}_p\mathrm{Br}(X) \to 0,$$

where $\pi\colon X \to \operatorname{Spec} F$, it follows that

$${}_p\mathrm{Br}(X)/{}_p\mathrm{Br}(F) \cong H^2(X, \mu_p)/H^2(F, \mu_p)$$

and hence the map ${}_p\mathrm{Br}(F) \to {}_p\mathrm{Br}(X)$ is also not surjective. Finally,

$${}_p\mathrm{Br}(F) = K_2(F)/pK_2(F), \qquad {}_p\mathrm{Br}(X) = H^0(X, \mathcal{K}_2/p\mathcal{K}_2),$$

and the exact sequence in §8

$$K_2(F)/pK_2(F) \to H^0(X, \mathcal{K}_2/p\mathcal{K}_2) \to N \cap \mu_p \to 0$$

shows that $N \supset \mu_p$.

Using the transfer map, one can generalize the previous theorem as follows:

(12.2) THEOREM. *Let D/F be a central simple algebra of squarefree degree r prime to* char F. *Then $c \in \operatorname{Nrd} D^*$ if and only if $\chi_c \cdot [D] = 0 \in H^3(F, \mu_r^{\otimes 2})$.*

§13. Hilbert's Theorem 90 for K_2 (a special case)

(13.1) PROPOSITION. *Suppose that $F \supset \mu_p$, and E/F is a cyclic extension of degree p. Then* $\ker(k_2(E) \xrightarrow{N} k_2(F)) = k_2(E)^{1-\sigma} + k_2(F)$ *($k_2 = K_2/pK_2$, and σ is a generator of* $\mathrm{Gal}(E/F)$).

PROOF. We let $V(F)$ denote $\ker(k_2(E) \xrightarrow{N} k_2(F))/k_2(E)^{1-\sigma} + k_2(F)$. If F' is an extension of the field F, then we define $V(F')$ using $E' = E \times_F F'$ in place of E. A standard argument, using the transfer map, shows that $V(F) \hookrightarrow V(F')$ if the degree $[F'\colon F]$ is finite and prime to p.

(13.1.1) *Let D/F be a cyclic division algebra of degree p over F, and let X be the corresponding Severi-Brauer variety. Then $V(F) \hookrightarrow V(F(X))$.*

PROOF. Suppose that $w \in K_2(E)$, $N(w) \in pK_2(F)$ and $w = (w')^{1-\sigma} + u_{E(X)}$, where $w' \in K_2(E(X))$ and $u \in K_2(F(X))$. Modifying w by an element in $K_2(F)$, we may assume that $N(w) = 0$, i.e., $0 = N(w) = pu$, and hence

$$u \in {}_pK_2(F(X)) = {}_pK_2(F) + \{\xi, F(X)^*\}.$$

Since $\{\xi, F(X)^*\} \subset K_2(E(X))^{1-\sigma}$, we may suppose that $u \in {}_pK_2(F)$. The local invariants of the element w' are σ-invariant; hence, for any point $x \in X$ of codimension 1 there exists an element $a_x \in F(x)^*$ such that $\partial_y(w') = a_x$, if $y \mid x$. It is clear that $\{a_x\}_{\operatorname{codim} x = 1}$ determines an element of the group $H^1(X, \mathcal{K}_2)$ which becomes zero in $H^1(X_E, \mathcal{K}_2)$. By the results in §8, there exists an element $u' \in K_2(F(X))$, such that $\partial_x(u') = a_x$. Changing w' to $w' - u'$, we find that $w' \in K_2(E)$ and $w \in K_2(E)^{1-\sigma} + K_2(F)$.

(13.1.2) It follows from (13.1.1) that there exists an extension $\tilde{F}/F$ such that the field $\tilde{F}$ does not have extensions of degree prime to p, $k_2(\tilde{F}) = 0$ and $V(F) \hookrightarrow V(\tilde{F})$. On the other hand, according to (10.6), $V(\tilde{F}) = 0$.

(13.2) COROLLARY. *Under the conditions of* (13.1), *if* ${}_pK_2(F) = \{\xi, F^*\}$, *then* (*see* (10.2))

$$\ker\Big(K_2(E) \xrightarrow{N} K_2(F)\Big) = K_2(E)^{1-\sigma}$$

If E/F is an arbitrary finite extension, we let $K_2'(E)$ denote the subgroup of $K_2(E)$ generated by the symbols $\{a, b\}$, where $a \in F^*$ and $b \in E^*$.

(13.3) LEMMA. *Let* $u \in K_2'(E)$; *then the following conditions are equivalent*:

1) $N_{E/F}(u) = 0$.

2) *There exist natural numbers*, r, N *and* m, *integers* n_j $(1 \leqslant j \leqslant N)$ *and* m_{ij} $(0 \leqslant i \leqslant m, 1 \leqslant j \leqslant N)$, *and elements* $t \in F^*$, $a_i \in F^*$ *and* $\beta_i \in E^*$ $(0 \leqslant i \leqslant m)$ *which satisfy the following equalities*:

a) $u = \sum_{i=0}^{m}\{a_i, \beta_i\}$.

b) $a_0^r = 1$.

c) $N_{E/F}(\beta_0) = \prod_{j=1}^{N}(1 - \prod_{k=0}^{m} a_k^{m_{kj}})^{m_{0j}n_j} \cdot t^r$.

d) $N_{E/F}(\beta_i) = \prod_{j=1}^{N}(1 - \prod_{k=0}^{m} a_k^{m_{kj}})^{m_{ij}n_j}$ *if* $i > 1$.

PROOF. 2) $\Rightarrow$ 1) We have $N_{E/F}(u) = \sum_0^m\{a_i, N_{\beta_i}\}$. We let $x_j \in F^*$ denote $\prod_0^m a_i^{m_{ij}}$; then $N\beta_i = \prod_1^N(1 - x_j)^{m_{ij}\cdot n_j}$ (up to a factor of t^r, if $i = 0$), and, since $a_0^r = 1$, we have

$$\sum_{i=0}^{m}\{a_i, N\beta_i\} = \sum_{i,j}\left\{a_i, (1-x_j)^{m_{ij}n_j}\right\} = \sum_j n_j\left\{\prod_i a_i^{m_{ij}}, 1 - x_j\right\} = \sum_j n_j\{x_j, 1 - x_j\} = 0.$$

1) $\Rightarrow$ 2). Now suppose that $Nu = 0$. Representing u as $\Sigma\{a_i', b_i'\}$, we see that there exist $x_j \in F^*$ and integers n_j such that

$$\sum_i a_i' \otimes N\beta_i' = \sum_j n_j(x_j \otimes 1 - x_j).$$

We choose a finitely generated subgroup $T \subset F^*$ containing a_i', $N\beta_i'$, x_j and $1 - x_j$ such that the preceding equality if fulfilled in the group $T \times F^*$. The group T is the direct sum of a free abelian group with basis $a_1, \ldots, a_m$ and the cyclic group of order r generated by a_0. We represent the elements x_j in the form $x_j = \prod_{i=0}^m a_i^{m_{ij}}$ and we replace the a_i' by their expressions in terms of the a_i. We then find that the element u has the form $u = \Sigma\{a_i, \beta_i\}$, where the equalities a)–d) are fulfilled.

(13.4) PROPOSITION. *Suppose that* $F \supset \mu_p$, *and* E/F *is a cyclic extension of degree* p. *Then*

$$\ker\Big(K_2'(E) \xrightarrow{N} K_2(F)\Big) \subset K_2(E)^{1-\sigma}.$$

PROOF. Let $u \in K_2'(E)$ and $N(u) = 0$; we find r, N, m, n_j, m_{ij}, t, a_i and b_i as in (13.3). Let $E = F(\sqrt[p]{c})$ and $\beta_i = \Sigma_{s=0}^{p-1} x_{is}(\sqrt[p]{c})^s$; then $N_{E/F}(\beta_i) = Q(x_{is}, c)$, where Q is some universal polynomial. We let F_0 denote the minimal subfield of F containing $\xi = \sqrt[p]{1}$ and a_0, and we consider the variety

$$V = \operatorname{Spec} F_0[C, T, A_i, X_{is}]_S \Big/ \left(Q(X_{is}, C) - \prod_{j=1}^{N}\left(1 - \prod_{k=0}^{m} A_k^{m_{kj}}\right)^{m_{ij}n_j} \cdot T^{\varepsilon_i}\right),$$

where S is the multiplicative system generated by the elements $A_1, \ldots, A_m$ and $1 - \prod_{k=0}^m A_k^{m_{kj}}$; $A_0 = a_0$, $\varepsilon_0 = r$, and $\varepsilon_i = 0$, if $i \geqslant 1$. Our elements determine a nonsingular point $v \in V$ and a natural map $V[\sqrt[p]{c}] \to V$ which is etale over v. We let $\tilde{F}$ denote the field

of rational functions on V. Since V is birationally isomorphic to the product of the Severi-Brauer varieties corresponding to the algebras

$$A_\xi\left(C, \prod_j \left(1 - \prod_{k=0}^{m} A_k^{m_{kj}}\right)^{m_{ij}n_j} \cdot T^{\varepsilon_i}\right)$$

over the field $F_0(C, T, A_i)$, it follows by (9.2) and Tate's theorem [18] that ${}_pK_2(\tilde{F}) = \{\xi, \tilde{F}^*\}$. Since

$$U = \sum_{i=0}^{m} \left\{A_i, \sum_s X_{is}\left(\sqrt[p]{C}\right)^i\right\} \in \ker\left(N_{\tilde{E}/\tilde{F}}\colon K_2(\tilde{E}) \to K_2(\tilde{F})\right),$$

it follows from (13.2) that $U \in K_2(\tilde{E})^{1-\sigma}$. Applying the specialization homomorphism, we see that $u \in K_2(E)^{1-\sigma}$.

PROPOSITION. *Let E/F be a cyclic extension of degree $p \neq$ char F. Then*

$$\ker\left(K_2(E) \xrightarrow{N} K_2(F)\right) = K_2(E)^{1-\sigma}.$$

PROOF. Using the transfer homomorphism, we reduce everything to the case when the field F does not have extensions of degree prime to p. But in this case $F \supset \mu_p$ and $K_2'(E) = K_2(E)$ (see [11]).

§14. Hilbert's Theorem 90 for K_2 (the general case)

(14.1) THEOREM. *Let E/F be a cyclic extension of degree n prime to* char F, *and let σ be a generator of* Gal(E/F). *Then*

$$\ker\left(K_2(E) \xrightarrow{N_{E/F}} K_2(F)\right) = K_2(E)^{1-\sigma}.$$

PROOF. Using the transfer homomorphism, we can reduce the general case to the case when $n = p^k$ and $F \supset \mu_p$. We let $V(F)$ denote

$$\ker\left(K_2(E) \xrightarrow{N_{E/F}} K_2(F)\right)/K_2(E)^{1-\sigma}.$$

Obviously, $V(F) \hookrightarrow V(F')$ if the degree $[F'\colon F]$ is finite and prime to p.

(14.1.1) Let D/F be a cyclic division algebra of degree p, and let X/F be the corresponding Severi-Brauer variety. Then $V(F) \hookrightarrow V(F(X))$.

The proof is carried out just as in (13.1.1).

(14.1.2) According to (14.1.1), we may assume that the field F does not have extensions of degree prime to p, and $K_2(F) = pK_2(F)$. The proof proceeds by induction on k. The case $k = 1$ was considered in (13.5). We consider an intermediate field L having degree p^{k-1} over F. By (10.6), this field satisfies the same conditions as F, and hence the transfer homomorphism $N_{E/F}\colon K_2(E) \to K_2(L)$ is surjective, since the group $K_2(L)$ is p-divisible. Suppose that $0 = N_{E/F}(u) = N_{L/F}(N_{E/L}(u))$. By the induction assumption, $N_{E/L}(u) = v^{\sigma-1}$ for some $v \in K_2(L)$, with (as shown above) $v = N_{E/L}(w)$, where $w \in K_2(E)$, i.e. $N_{E/L}(u) = N_{E/L}(v^{\sigma-1})$. Finally, $u \cdot w^{1-\sigma} \in \ker N_{E/F} = K_2(E)^{\tau-1} \subset K_2(E)^{\sigma-1}$ ($\tau = \sigma^{p^{k-1}}$), and hence $u \in K_2(E)^{\sigma-1}$.

(14.2) THEOREM. *If $F \supset \mu_n$, then ${}_nK_2(F) = \{\mu_n, F^*\}$.*

PROOF. It is easy to reduce the general case to the case when $n = p^k$, where p is a prime. The proof uses induction on k. The case $k = 1$ follows from (10.4) and (14.1) or (13.5). Let

η be a primitive p^kth root of unity such that $\eta^{p^{k-1}} = \xi$. If $p^k u = 0$, then $p^{k-1}u \in \{\xi, F^*\}$, and hence $p^{k-1}u = \{\xi, f\} = p^{k-1}\{\eta, f\}$. By the induction assumption, $u - \{\eta, f\} \in \{\mu_{p^{k-1}}, F^*\}$; therefore, $u \in \{\mu_{p^k}, F^*\}$.

§15. An exact cohomology sequence

From the Hilbert Theorem 90 for K_2 in (14.1) and the fundamental theorem (11.5) it is easy to deduce

(15.1) PROPOSITION. *Suppose that* $(n, \operatorname{char} F) = 1$, E/F *is a cyclic extension of degree* n, *and* σ *is a generator of* $\operatorname{Gal}(E/F)$. *Then the following sequence is exact*:

$$H^2\big(F, \mu_n^{\otimes 2}\big) \oplus H^2\big(E, \mu_n^{\otimes 2}\big) \xrightarrow{(\operatorname{Res}, 1-\sigma)} H^2\big(E, \mu_n^{\otimes 2}\big) \xrightarrow{N_{E/F}} H^2\big(F, \mu_n^{\otimes 2}\big).$$

(15.2) PROPOSITION. *Suppose that* G *is a group and* $H \subset G$ *is a normal subgroup such that* $G/H \cong \mathbf{Z}/n\mathbf{Z}$; $\chi_0 \in H^1(G/H, \mathbf{Z}/n\mathbf{Z})$ *and* $\theta_0 \in H^2(G/H, \mathbf{Z}/n\mathbf{Z})$ *are generators of the respective groups, and* $\chi \in H^1(G, \mathbf{Z}/n\mathbf{Z})$ (*respectively* $\theta \in H^2(G, \mathbf{Z}/n\mathbf{Z})$) *are their images under the inflation map. Let* σ *be a generator of the group* G/H. *If* M *is a* $\mathbf{Z}/n\mathbf{Z}[G]$*-module which is free over* $\mathbf{Z}/n\mathbf{Z}$, *then for every* $i \geqslant 1$ *there exists a canonical isomorphism*

$$\ker\big(N_{G/H}\colon H^i(H, M) \to H^i(G, M)\big)/H^i(H, M)^{1-\sigma} + \operatorname{Res}\big(H^i(G, M)\big)$$
$$\cong \ker\big(\operatorname{Res}\colon H^{i+1}(G, M) \to H^{i+1}(H, M)\big)/\chi \cdot H^i(G, M) + \theta \cdot H^{i-1}(G, M).$$

PROOF. Let V denote the kernel of the natural map $M[G/H] \to M$; then we have the two exact sequences of G-modules

$$0 \to V \to M[G/H] \to M \to 0,$$
$$0 \to M \xrightarrow{1+\sigma+\cdots+\sigma^{n-1}} M[G/H] \to V \to 0$$

and hence the exact cohomology sequences

$$H^{i-1}(G, M) \xrightarrow{\delta} H^i(G, V) \to \ker\big(N_{G/H}\colon H^i(H, M) \to H^i(G, M)\big) \to 0,$$

$$H^i(H, M) \to H^i(G, V) \xrightarrow{\delta} \ker\big(\operatorname{Res}\colon H^{i+1}(G, M) \to H^{i+1}(H, M)\big) \to 0.$$

The composition $H^{i-1}(G, M) \xrightarrow{\delta} H^i(G, V) \xrightarrow{\delta} H^{i+1}(G, M)$ coincides (up to a sign) with the connecting homomorphism for the exact sequence

$$0 \to M \to M[G/H] \to M[G/H] \to M \to 0$$

and thus coincides with multiplication by θ. We thereby obtain the exact sequence

$$H^i(H, M) \to H^i(G, V)/\delta\big(H^{i-1}(G, M)\big) = \ker N \to \ker \operatorname{Res}/\theta \cdot H^{i-1}(G, M) \to 0.$$

It is not hard to show that the composition $H^i(H, M) \to H^i(G, V) \to H^i(H, M)$ coincides with $\sigma - 1$; hence, we have a canonical isomorphism

$$\ker\big(N_{G/H}\colon H^i(H, M) \to H^i(G, M)\big)/H^i(H, M)^{\sigma-1}$$
$$\cong \ker\big(\operatorname{Res}\colon H^{i+1}(G, M) \to H^{i+1}(H, M)\big)/\theta \cdot H^{i-1}(G, M).$$

To complete the proof, it suffices to note that this isomorphism takes the natural map $\operatorname{Res}\colon H^i(G, M) \to \ker N$ to $\chi\colon H^i(G, M) \to \ker \operatorname{Res}$.

(15.3) COROLLARY. *Under the conditions of* (15.1), *the map*

$$\ker\Big(H^3\big(F,\mu_n^{\otimes 2}\big)\xrightarrow{\mathrm{Res}}H^3\big(E,\mu_n^{\otimes 2}\big)\Big)$$

coincides with $\chi\cdot H^2(F,\mu_n^{\otimes 2})+\theta\cdot H^1(F,\mu_n^{\otimes 2})$. *If* $F\supset\mu_n$, *then this group coincides with* $\chi\cdot H^2(F,\mu_n^{\otimes 2})$.

PROOF. The first assertion is obvious. If $F\supset\mu_n$, then we can identify the group $H^2(F,\mathbf{Z}/n\mathbf{Z})$ with $H^2(F,\mu_n)={}_n\mathrm{Br}(F)$. In that case $\theta\in\ker({}_n\mathrm{Br}(F)\to{}_n\mathrm{Br}(E))$; hence $\theta=\chi\cup\lambda$ for some $\lambda\in H^1(F,\mu_n)$, i.e., θ is divisible by χ.

(15.4) Using the transfer map, we can prove the following fact: *If* n *is squarefree, then* θ *is divisible by* χ (even in the case when $F\not\supset\mu_n$). This is apparently true for any n.

(15.5) PROPOSITION. *Under the conditions of* (15.2), *suppose that* $n=p$ *is a prime. Then the groups*

$$\ker(\mathrm{Res}\colon H^{k+1}(G,M)\to H^{k+1}(H,M))/\chi\cdot H^k(G,M)$$

and

$$\ker\Big(\bigcup\chi\colon H^k(G,M)\to H^{k+1}(G,M)\Big)/N_{G/H}\big(H^k(H,M)\big)$$

have filtrations whose successive quotients are isomorphic.

PROOF. Let $P_i=\mathbf{Z}/p\mathbf{Z}\oplus\cdots\oplus\mathbf{Z}/p\mathbf{Z}$ (with i terms, $i=1,\dots,p$), where the action of the group G/H is given by the formula $\sigma(x_1,\dots,x_i)=(x_1+x_2,\dots,x_{i-1}+x_i,x_i)$. In particular, $P_1=\mathbf{Z}/p\mathbf{Z}$ and $P_p=\mathbf{Z}/p\mathbf{Z}[G/H]$. We set $M_i=M\otimes P_i$; if $i\leqslant j$, we have the G-homomorphisms $\varphi_{ij}\colon P_i\to P_j$ and $\psi_{ji}\colon P_j\to P_i$, which are defined by

$$\varphi_{ij}(x_1,\dots,x_i)=(x_1,\dots,x_i,0,\dots,0),\qquad \psi_{ji}(x_1,\dots,x_j)=(x_{j-i+1},\dots,x_j).$$

We shall use the same letters φ_{ij} and ψ_{ji} to denote the induced homomorphisms $M_i\to M_j$ and $M_j\to M_i$. The following properties are obvious.

1) If $i\leqslant j\leqslant k$, then $\varphi_{jk}\cdot\varphi_{ij}=\varphi_{ik}$ and $\psi_{ji}\cdot\psi_{kj}=\psi_{ki}$.

2) The sequence $0\to P_i\xrightarrow{\varphi}P_j\xrightarrow{\psi}P_{j-i}\to 0$ is exact.

3) The compositions $P_i\xrightarrow{\varphi}P_j\xrightarrow{\psi}P_i$ and $P_j\xrightarrow{\psi}P_i\xrightarrow{\varphi}P_j$ coincide with $(\sigma-1)^{j-i}$.

The sequence of G-homomorphisms $M_p\xrightarrow{\psi}M_{p-1}\to\cdots\to M_1=M$ determines a filtration on $H^k(G,M)$ given by

$$H^k(G,M)_i=\mathrm{im}\Big(H^k(G,M_i)\xrightarrow{\psi_*}H^k(G,M)\Big).$$

We have the obvious formulas

$$H^k(G,M)_p=\mathrm{im}\,N_{G/H}\big(H^k(H,M)\big),$$

$$H^k(G,M)_2=\mathrm{im}\big(H^k(G,M_2)\to H^k(G,M)\big)=\ker\Big(\bigcup\chi\colon H^k(G,M)\to H^{k+1}(G,M)\Big).$$

The second group in the statement of the proposition obviously coincides with $H^k(G,M)_{2/p}$. We compute the successive quotients in our filtration:

$$H^k(G,M)_{i/i+1}=H^k(G,M_i)/\big((\psi_{i+1,i})_*\big(H^k(G,M_{i+1})\big)+(\varphi_{i-1,i})_*\big(H^k(G,M_{i-1})\big)\big).$$

From the exact sequence of G-modules

$$0 \to M_i \xrightarrow{(\varphi_{i,i+1}, -\psi_{i,i-1})} M_{i+1} \oplus M_{i-1} \xrightarrow{(\psi_{i+1,i}, \varphi_{i-1,i})} M_i \to 0$$

we find that

$$H^k(G, M)_{i/i+1} = \ker\bigl(\varphi_*\colon H^{k+1}(M_i) \to H^{k+1}(M_{i+1})\bigr) \cap \ker\bigl(\psi_*\colon H^{k+1}(M_i) \to H^{k+1}(M_{i-1})\bigr).$$

We consider the exact sequence of G-modules $M = M_1 \xrightarrow{\varphi} M_2 \to \cdots \to M_p$. It determines a filtration on $H^{k+1}(G, M)$ as follows:

$$H^{k+1}(G, M)^i = \ker\bigl(H^{k+1}(G, M) \to H^{k+1}(G, M_i)\bigr).$$

In particular,

$$H^{k+1}(G, M)^2 = \chi \cdot H^k(G, M),$$
$$H^{k+1}(G, M)^p = \ker(\mathrm{Res}\colon H^{k+1}(G, M) \to H^{k+1}(H, M)),$$

and the first group in the statement of the proposition coincides with $H^{k+1}(G, M)^{p/2}$. The successive quotients in this filtration have the form

$$H^{k+1}(G, M)^{i+1/i} = \mathrm{im}(\varphi_{1,i})_* \cap \ker(\varphi_{i,i+1})_* = \ker(\psi_{i,i-1})_* \cap \ker(\varphi_{i,i+1})_* = H^k(G, M)_{i/i+1}.$$

(15.6) THEOREM. *Let E/F be a cyclic extension of prime degree p. Then the following sequence is exact:*

$$H^2\bigl(F, \mu_p^{\otimes 2}\bigr) \oplus H^2\bigl(E, \mu_p^{\otimes 2}\bigr) \xrightarrow{(\mathrm{Res}, 1-\sigma)} H^2\bigl(E, \mu_p^{\otimes 2}\bigr) \xrightarrow{N_{E/F}} H^2\bigl(F, \mu_p^{\otimes 2}\bigr) \xrightarrow{\chi} H^3\bigl(F, \mu_p^{\otimes 2}\bigr) \xrightarrow{\mathrm{Res}} H^3\bigl(E, \mu_p^{\otimes 2}\bigr).$$

(15.7) REMARKS. a) In the statement of the theorem, p can be replaced by any squarefree number. Apparently, the corresponding assertion is true for all cyclic extensions.

b) If $p = 2$, then our exact sequence coincides with a part of the well-known infinite exact sequence for a quadratic extension.

§16. The structure of central simple algebras

The following assertion follows immediately from the fundamental theorem.

(16.1) THEOREM. *Let D/F be a central simple algebra of exponent n. If $F \supset \mu_n$, then the algebra D is similar to a tensor product of cyclic algebras of exponent n; in particular, D has an abelian splitting field. In the general case, D has a solvable splitting field.*

(16.2) COROLLARY. *If the field F does not have abelian extensions, then* $\mathrm{Br}(F) = 0$.

It seems likely that (16.1) holds even if $F \not\supset \mu_n$. We can prove this in two cases.

(16.3) LEMMA. *Suppose that E/F is a quadratic extension, and $E \supset \mu_n$. Then the transfer map $N_{E/F}\colon H^2(E, \mu_n) \to H^2(F, \mu_n)$ takes the classes of cyclic algebras to sums of classes of cyclic algebras.*

PROOF. It is well known that the group $K_2(E)$ is generated by the symbols $\{\beta, a\}$, where $a \in F^*$ and $\beta \in E^*$. We let $\varphi_\beta \in H^1(E, \mathbf{Z}/n\mathbf{Z})$ and $\chi_a \in H^1(F, \mu_n)$ denote the corresponding cohomology classes; then $[A_\xi(\beta, a)] = \varphi_\beta \cup \operatorname{Res} \chi_a$, and hence $N_{E/F}[A_\xi(\beta, a)] = N_{E/F}(\varphi_\beta) \cup \chi_a$ is the class of a cyclic algebra.

(16.4) COROLLARY. *The group* ${}_3\mathrm{Br}(F)$ *is generated by the classes of cyclic algebras.*

(16.5) LEMMA. *If* $[D] \in {}_2\mathrm{Br}(F)$, *then the following conditions are equivalent*:
a) $[D] \in 2\,\mathrm{Br}(F)$.
b) $[d] \in N_{F(\mu_4)/F}({}_2\mathrm{Br}(F(\mu_4)))$.

PROOF. If $\xi \in \mu_4$ is a primitive root of one, then $[A_{-1}(\alpha, \beta)] = 2[A_\xi(\alpha, \beta)]$, and hence ${}_2\mathrm{Br}(F(\mu_4)) \subset 2\,\mathrm{Br}(F(\mu_4))$, i.e., b) $\Rightarrow$ a).

Conversely, consider the exact sequences of G-modules ($G = \mathrm{Gal}(F)$)

$$E_1\colon 0 \to \mu_2 \to \mu_4 \to \mu_2 \to 0,$$
$$E_2\colon 0 \to \mathbf{Z}/2\mathbf{Z} \to \mathbf{Z}/4\mathbf{Z} \to \mathbf{Z}/2\mathbf{Z} \to 0,$$
$$E_3\colon 0 \to \mu_2 \to \mu_2[G/H] \to \mu_2 \to 0$$

($H = \mathrm{Gal}(F(\mu_4))$). If we identify μ_2 and $\mathbf{Z}/2\mathbf{Z}$, then in $\mathrm{Ext}^1_G(\mu_2, \mu_2)$ we have $[E_1] = [E_2] + [E_3]$; consequently, $\delta_1 = \delta_2 + \delta_3$, where δ_i is the connecting homomorphism corresponding to the sequence E_i ($i = 1, 2, 3$). Condition a) is equivalent to the equality $\delta_1([D]) = 0$, and condition b) is equivalent to $\delta_3([D]) = 0$. Hence, it suffices to prove that $\delta_2([D]) = 0$.

(16.5.1) LEMMA. *For the group* $G = \mathrm{Gal}(F)$, *the restriction of the connecting homomorphism* $\delta_2\colon H^2(G, \mathbf{Z}/2\mathbf{Z}) \to H^3(G, \mathbf{Z}/2\mathbf{Z})$ *to the image of the map* $H^1(G, \mathbf{Z}/2\mathbf{Z}) \times H^1(G, \mathbf{Z}/2\mathbf{Z}) \to H^2(G, \mathbf{Z}/2\mathbf{Z})$ *is zero.*

The proof is simple, so we omit it.

(16.6) PROPOSITION. *The group* ${}_4\mathrm{Br}(F)$ *is generated by the classes of cyclic algebras.*

PROOF. Let $x \in {}_4\mathrm{Br}(F)$; according to (16.5), there exists $y \in {}_2\mathrm{Br}(F(\mu_4))$ such that $2x = N(y)$. We can replace y by $2z$, where z is a sum of classes of cyclic 4-algebras. By (16.3), Nz is also a sum of classes of cyclic 4-algebras; but then the element $x - Nz$ has period 2, and so is a sum of classes of cyclic 2-algebras.

(16.7) PROPOSITION. *If* F *is a* C_2*-field, then every algebra of exponent* 2 *or* 3 *is cyclic.*

PROOF. Since every algebra of degree 2 or 3 is cyclic by Albert's theorem [8], it suffices to prove that any two algebras (of exponent 2 or 3) have a common splitting field, i.e., there exist elements $x_1 \in D_1 - F$ and $x_2 \in D_2 - F$ with the same reduced characteristic polynomial. In the case of quaternion algebras, this problem reduces to solving a system of one linear and one quadratic equation with 8 variables; such a system is solvable over a C_2-field. In the case of algebras of degree 3, we arrive at a system of three equations of degree 1, 2 and 3 with 18 variables; such a system is also solvable.

(16.8) COROLLARY. *If a central simple algebra over a* C_2*-field has exponent* $n = 2^a \cdot 3^b$, *then the degree of this algebra is equal to its exponent.*

PROOF. By Albert's theorem [8], it suffices to consider the cases $n = 2$ and $n = 3$.

(16.9) REMARK. The fact that every algebra of exponent 2 over a C_2-field is a quaterion algebra was proved earlier by V. I. Yanchevskiĭ. The observation that the product of cyclic algebras of degree 3 is similar to a cyclic algebra of degree 3 is also due to him.

(16.10) PROPOSITION. *Over a C_n-field, every algebra of exponent 2 is similar to a product of no more than $2^{n-1} - 1 = t$ cyclic algebras; hence, the degree of a division algebra of exponent 2 cannot exceed 2^t.*

PROOF. It suffices to prove that every element of $K_2(F)/2K_2(F)$ is a sum of no more than $2^{n-1} - 1$ symbols. Because of the isomorphism $K_2(F)/2K_2(F) \cong I^2/I^3$ in [19], where I is the ideal of even-dimensional quadratic forms in the Witt ring $W(F)$, it is sufficient to show that every anisotropic form $q \in I^2$ is congruent modulo the ideal I^3 to a sum of no more than $2^{n-1} - 1$ Pfister 2-forms. Since F is a C_n-field, it follows that $\dim q \leqslant 2^n$, and

$$\begin{aligned} q &= \langle a_1, a_2, \ldots, a_{2^n} \rangle \\ &= \langle a_1, a_2, a_3, a_1a_2a_3 \rangle + \langle -a_1a_2a_3, a_4, a_5, -a_1a_2a_3a_4a_5 \rangle + \cdots \\ &\equiv \langle\langle -a_1a_2, -a_1a_3 \rangle\rangle + \langle\langle a_1a_2a_3a_4, a_1a_2a_3a_5 \rangle\rangle + \cdots \end{aligned}$$

(the number of terms is equal to $2^{n-1} - 1$).

§17. Division algebras over two-dimensional fields

(17.1) LEMMA. *Let D/F be a central simple algebra of squarefree degree n. Suppose that $(n, \operatorname{char} F) = 1$, and there exists a cyclic splitting field E/F for the algebra D such that the map θ_E: $K_2(E) \to K_2(D)$ is surjective. Then the reduced norm map* Nrd: $K_2(D) \to K_2(F)$ *is injective, and its image coincides with the image of the homomorphism $N_{E/F}$: $K_2(E) \to K_2(F)$.*

PROOF. We consider the following commutative diagram:

$$\begin{array}{ccc} K_2(L) & \xrightarrow{\theta_L} & K_2(D) \\ {\scriptstyle N_{L/F}}\searrow & & \swarrow{\scriptstyle \mathrm{Nrd}} \\ & K_2(F) & \end{array}$$

It suffices to show that $\ker(N_{E/F}) \subset \ker \theta_E$; but this follows easily from the Hilbert Theorem 90 for K_2. In fact, replacing E by a subfield, we may assume that $([E: F], \operatorname{char} F) = 1$. By Hilbert's Theorem 90, we have $\ker(N_{E/F}) = K_2(E)^{1-\sigma}$, where σ is a generator of $\mathrm{Gal}(E/F)$; but $\theta_E\sigma = \theta_E$ (§1), and hence $K_2(E)^{1-\sigma} \subset \ker \theta_E$.

(17.1.1) EXAMPLE. Let $F = \mathbf{R}$ be the field of real numbers, and let D be the classical quaternion algebra. Since $\mathbf{C}$ is the only finite splitting field for D, it follows from the results of §5 that the map $\theta_{\mathbf{C}}$ is surjective. Consequently, Nrd maps $K_2(D)$ isomorphically onto the subgroup $N_{\mathbf{C}/\mathbf{R}}(K_2(\mathbf{C})) \subset K_2(\mathbf{R})$, which coincides with the subgroup generated by the symbols $\{x, y\}$, where $x, y > 0$, i.e., we have obtained the Alperin-Dennis theorem [9].

(17.2) PROPOSITION. *Let* c.d. $F \leqslant 2$, *and let E/F be an extension of degree prime to* char F. *Then the map $N_{E/F}$: $K_2(E) \to K_2(F)$ is surjective.*

PROOF. Let n denote $[E: F]$; since $N_{E/F} \supset nK_2(F)$, it follows that we can replace K_2 by $k_2 = K_2/nK_2$. But $k_2(F) = H^2(F, \mu_n^{\otimes 2})$, and the map $H^2(E, \mu_n^{\otimes 2}) \xrightarrow{N_{E/F}} H^2(F, \mu_n^{\otimes 2})$ is surjective, since its cokernel is isomorphic to a subgroup of some three-dimensional cohomology group.

(17.3) THEOREM. *If* c.d. $F \leqslant 2$ *and* D/F *is a central simple algebra of squarefree degree prime to* char F, *then* Nrd: $K_2(D) \xrightarrow{\sim} K_2(F)$

PROOF. The general case is easily reduced to the case when the degree of D is p and the field F does not have extensions of degree prime to p. Let E and E' be splitting fields of D; we consider the following commutative diagram:

$$\begin{array}{ccc} & K_2(E') & \\ {\scriptstyle N_{EE'/E'}}\nearrow & & \searrow{\scriptstyle \theta_{E'}} \\ K_2(EE') & \xrightarrow{\theta_{EE'}} & K_2(D) \\ {\scriptstyle N_{EE'/E}}\searrow & & \nearrow{\scriptstyle \theta_E} \\ & K_2(E) & \end{array}$$

Surjectively of the transfer maps implies equality of the images $\operatorname{im}\theta_E = \operatorname{im}\theta_{EE'} = \operatorname{im}\theta_{E'}$; consequently, the results in §5 show that the map θ_E is surjective. We can now take for E the maximal subfield in D which is cyclic over F, and the proof is completed by applying (17.1).

(17.4) THEOREM. *Let* F *be a local or global field, and let* D/F *be a division algebra of squarefree degree prime to* char F. *Then the following sequence is exact*:

$$0 \to K_2(D) \xrightarrow{\text{Nrd}} K_2(F) \to \coprod_S \mathbf{Z}/2\mathbf{Z} \to 0,$$

where S *is the set of real places* v *of* F *such that the algebra* $D_v = D \otimes_F F_v$ *is nontrivial.*

PROOF. Standard arguments using the transfer map reduce the proof to the case when the algebra has degree p. If $p \neq 2$, then we may assume that the field F does not have real places, and hence that c.d. $F = 2$ (see [5]). Now suppose that $p = 2$, i.e., D is a quaternion algebra. We can find a quadratic splitting field E for D with the property that if v is a real place of F such that D_v is trivial, then $E_v = \mathbf{R} \times \mathbf{R}$. Using the description of the Brauer group of a global field (see [10], [23]), we easily see that, for any splitting field E' of D, the map ${}_2\mathrm{Br}(EE') \xrightarrow{N} {}_2\mathrm{Br}(E')$, and hence also $K_2(EE') \xrightarrow{N} K_2(E')$, are surjective. As in the proof of (17.3), we find that the map θ_E is surjective; thus, Nrd: $K_2(D) \hookrightarrow K_2(F)$ and

$$K_2(F)/K_2(D) = K_2(F)/N_{E/F}(K_2(E)) = {}_2\mathrm{Br}(F)/N_{E/F}({}_2\mathrm{Br}(E)) = \coprod_S \mathbf{Z}/2\mathbf{Z}.$$

(17.5) REMARK. Theorem (17.4) improves upon some results of Rehmann and Stuhler [22], but, unfortunately, only in the case of algebras of squarefree degree. It seems that the sequence (17.4) is exact for any algebra D.

A field F is called a C_2'-*field* [5] if the map Nrd: $D^* \to F^*$ is surjective for any division algebra D/F, and if the same holds for any finite extension E/F (i.e., the map Nrd: $D^* \to E^*$ is surjective for an algebra D/E).

(17.6) PROPOSITION. *Let* D/F *be a central simple algebra of squarefree degree prime to* char F. *Suppose that the map* Nrd: $D_E^* \to E^*$ *is surjective for any finite extension* E/F (*for example,* F *is a* C_2'-*field*). *Then* Nrd: $K_2(D) \to K_2(F)$ *is an isomorphism.*

PROOF. As usual, we may assume that the algebra D has prime degree p and the field F does not have extensions of degree prime to p. Under our assumptions we have Nrd: $K_1(D) \xrightarrow{\sim} F^*$; consequently, we have the multiplication map $F^* \times F^* = K_1(F) \times K_1(D) \to K_2(D)$, which takes the element $a \times b$ to $c(a, \beta)$, where $\beta \in D^*$ and $\operatorname{Nrd} \beta = b$. Rehmann and Stuhler [22] proved that, under the assumptions of the proposition, the map $F^* \times F^* \to K_2(D)$ is a Steinberg symbol, and so gives a homomorphism ψ: $K_2(F) \to K_2(D)$, which is surjective in view of the results in §5. It remains to note that the composition $K_2(F) \to K_2(D) \xrightarrow{\text{Nrd}} K_2(F)$ is the identity.

§18. Cycles of codimension two

From results of Bloch [13] and the fundamental theorem of the present paper it follows that, if X is a smooth variety over a field F and $(n, \operatorname{char} F) = 1$, then

$$H^1(X, \mathcal{K}_2/n\mathcal{K}_2) \cong NH^3(X_{\text{et}}, \mu_n^{\otimes 2}),$$

where NH^3 is the subgroup consisting of cohomology classes which vanish under restriction to the generic point of X; and there exists an exact sequence

$$H^1(X, \mathcal{K}_2) \xrightarrow{n} H^1(X, \mathcal{K}_2) \to NH^3(X_{\text{et}}, \mu_n^{\otimes 2}) \to {}_n\mathrm{Ch}^2(X) \to 0.$$

(18.1) COROLLARY. *If the field F is algebraically closed or a finite field, and the variety X is projective, then the group ${}_n\mathrm{Ch}^2(X)$ is finite for any n prime to* char F.

(18.2) From the definitions of the maps in the exact sequence it is not hard to deduce that for any integer m prime to char F, the following diagram is commutative:

$$\begin{array}{ccccccccc} H^1(X, \mathcal{K}_2) & \xrightarrow{n} & H^1(X, \mathcal{K}_2) & \to & NH^3(X_{\text{et}}, \mu_n^{\otimes 2}) & \to & {}_n\mathrm{Ch}^2(X) & \to & 0 \\ \| & & \downarrow m & & \downarrow & & \supset & & \\ H^1(X, \mathcal{K}_2) & \xrightarrow{nm} & H^1(X, \mathcal{K}_2) & \to & NH^3(X_{\text{et}}, \mu_{nm}^{\otimes 2}) & \to & {}_{nm}\mathrm{Ch}(X) & \to & 0 \end{array}$$

in which the map of cohomology groups is induced by the imbedding $\mu_n^{\otimes 2} \hookrightarrow \mu_{nm}^{\otimes 2}$. Now let n run through the set of powers of a fixed prime l. Taking the limit of the corresponding exact sequences, we obtain

(18.3) COROLLARY. *If $l \neq$ char F, then there exists a functorial exact sequence*

$$H^1(X, \mathcal{K}_2) \to H^1(X, \mathcal{K}_2)_l \to NH^3(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2)) \to \mathrm{Ch}^2(X)(l) \to 0,$$

where $H^1(X, \mathcal{K}_2)_l$ is the localization with respect to the multiplicative system $\{1, l, l^2, \dots\}$, and $\mathrm{Ch}^2(X)(l)$ is the l-torsion subgroup of $\mathrm{Ch}^2(X)$.

(18.4) We now suppose that the variety X is projective and the field F is algebraically closed. In this situation Bloch [12] constructed a map

$$\lambda^2\colon \mathrm{Ch}^2(X)(l) \to H^3(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2));$$

a direct calculation shows that the composition

$$NH^3(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2)) \to \mathrm{Ch}^2(X)(l) \to H^3(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2))$$

is an imbedding. This gives us additional information:

a) $NH^3(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2)) \xrightarrow{\sim} \mathrm{Ch}^2(X)(l)$.

b) $H^1(X, \mathcal{K}_2) \to H^1(X, \mathcal{K}_2)_l$ is an epimorphism, i.e., $H^1(X, \mathcal{K}_2)$ is the direct sum of the group $H^1(X, \mathcal{K}_2)(l)$ and a uniquely l-divisible subgroup.

c) $H^1(X, \mathcal{K}_2)(l)/l^k H^1(X, \mathcal{K}_2)(l) = H^2(X, \mathbf{Q}_l/\mathbf{Z}_l(2))/l^k H^2(X, \mathbf{Q}_l/\mathbf{Z}_l(2))$.

In order to prove the last formula, we must verify that

$$\ker\left(H^3\left(X_{\text{et}}, \mu_{l^k}^{\otimes 2}\right) \to H^3\left(X_{\text{et}}, \mathbf{Q}_l/\mathbf{Z}_l(2)\right)\right) \subset NH^3\left(X_{\text{et}}, \mu_{l^k}^{\otimes 2}\right),$$

i.e., that

$$H^3\left(F(X), \mu_{l^k}^{\otimes 2}\right) \hookrightarrow H^3\left(F(X), \mathbf{Q}_l/\mathbf{Z}_l(2)\right).$$

But the kernel of this homomorphism coincides with $H^2(F(X), \mathbf{Q}_l/\mathbf{Z}_l(2))/l^k$, so that it suffices to prove that the group $H^2(F(X), \mathbf{Q}_l/\mathbf{Z}_l(2))$ is l-divisible. But it immediately follows from the fundamental theorem that

$$H^2(F(X), \mathbf{Q}_l/\mathbf{Z}_l(2)) = K_2(F(X))_l/K_2(F(X))$$

and this group is l-divisible.

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